Chapter 2 - Wireless Transmission

Lecture Mobile Communication

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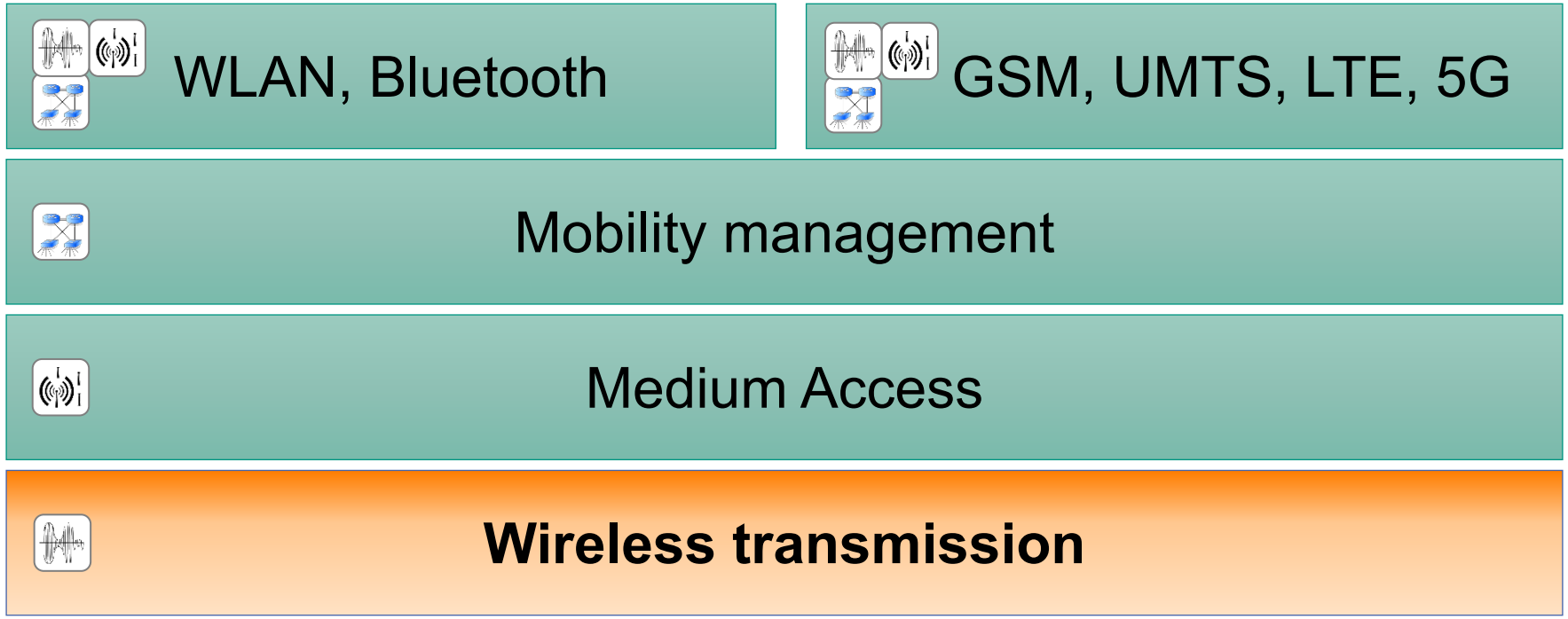
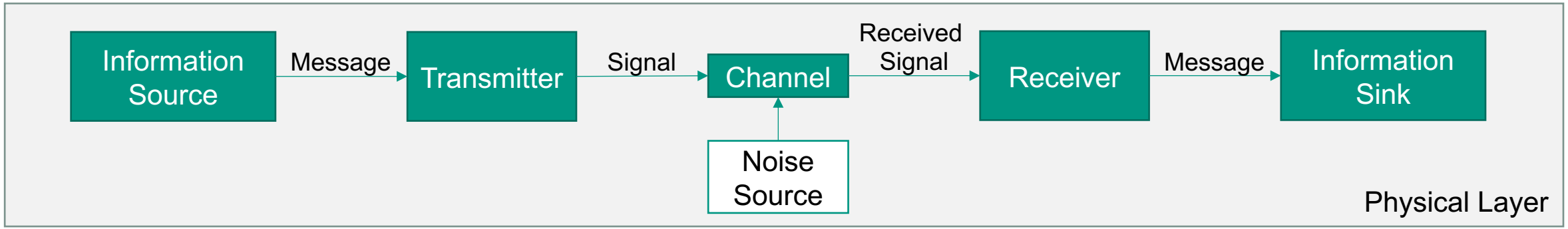
Structure


WLAN, Bluetooth
GSM, UMTS, LTE, 5G
Mobility management
Medium Access
Wireless transmission

Signals, Electromagnetic Waves and Modulation

Recap: Shannon´s Communication Model

- Information Source
  - Provides a message (bit sequence) to the transmitter
- Transmitter
  - Converts the bit sequence into a signal and sends it over the transmission medium or channel
- Receiver
  - Reconstructs the bit sequence from the received signal and passes it to the information sink.
- Information Sink
  - Receives the message.
- Noise Source
  - Introduces distortions into the transmitted signal



Signals

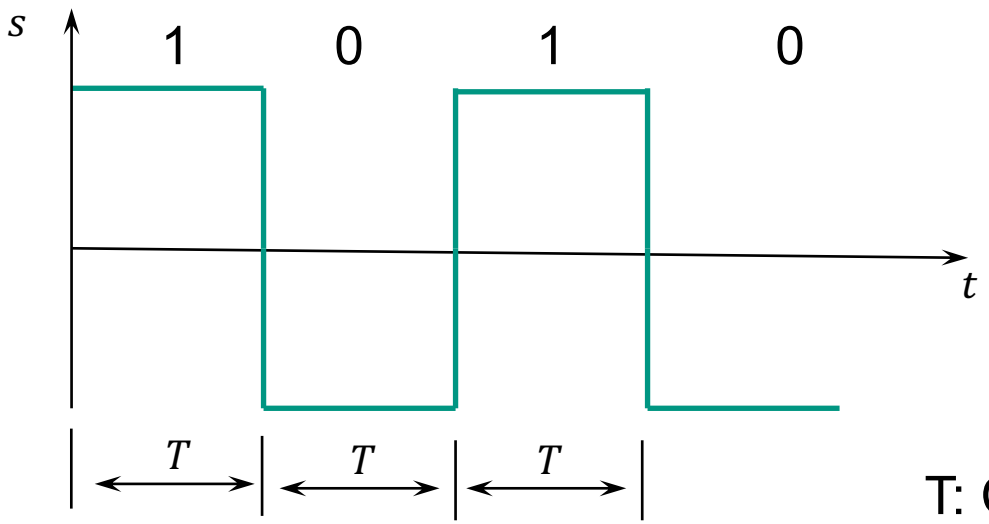
- Intuitively, a signal is the physical representation of data
  - Used for transmission of information over the channel

- Definition: Signal
  - Mathematical function of at least one independent variable
  - Varies as a function of time
  - At time t, the signal has the value $s(t)$

- Classification of Signals
  - Continuous or discrete in time
  - Continuous or discrete in values (often referred to as amplitude)

Digital Signals

- Definition
  - A digital signal is <u>discrete in time and amplitude</u>
  - We consider only isochronous digital signals
    - Fixed step duration T (clock period)
    - Changes in (signal) value occur only at clock period
  - Example of an isochronous digital signal



T: Clock period

Analog Signals

- **Definition**
  - An analog signal is <u>continuous in time and amplitude</u>
  - A periodic analog signal has a period T
    - Signal value $s(t)$ repeats after T time units:

$$s(t+T) = s(t), \qquad -\infty < t < \infty$$

Sinusoidal Signals

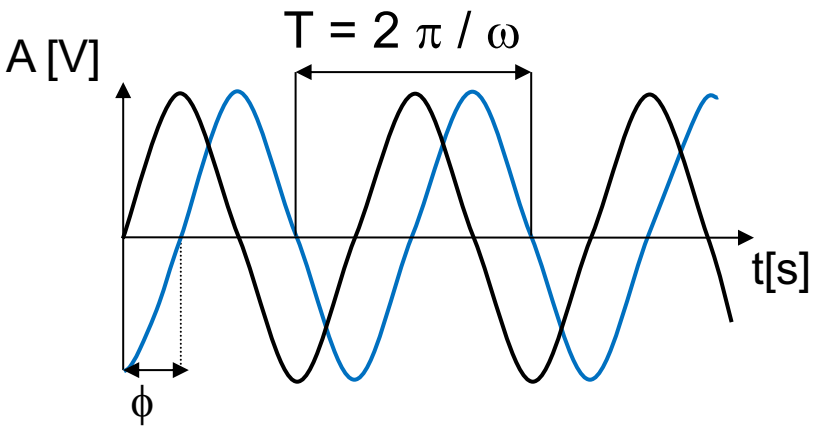
- Mobile communications are often based on radio transmissions
  - Radio waves generated by oscillations in the electromagnetic field
  - Waves can be modeled by sinusoidal functions (sin, cos, …)

→ We consider sinusoidal signals

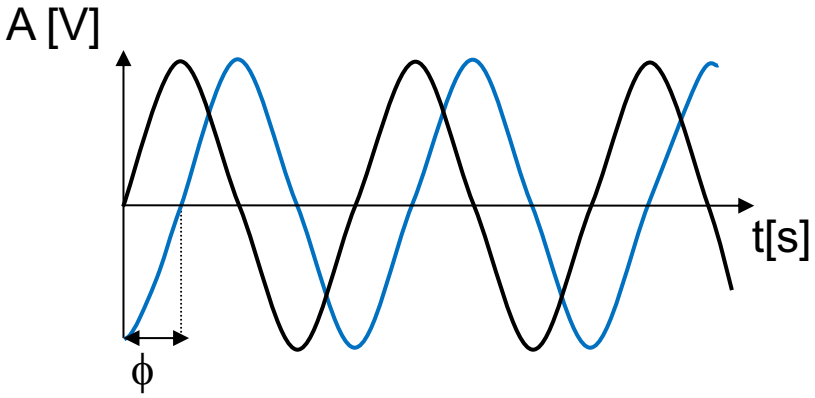
- Special case of a periodic analogous signal
- Example:

$$s(t) = A\sin(2\pi f t + \phi) = A\sin(\omega t + \phi)$$

- Parameters:
  - Amplitude A,
  - Period T,
  - Frequency $f = \frac{1}{T}$,
  - Angular frequency $\omega = 2\pi f$
  - Phase shift ϕ

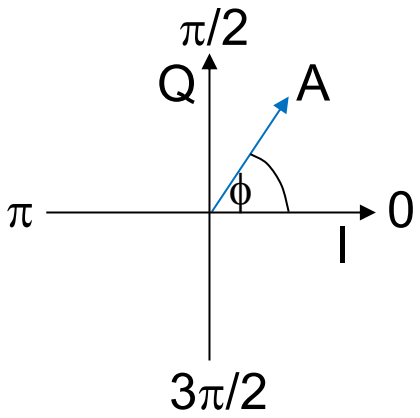


(Graphical) Representation of sinusoidal signals



Amplitude spectrum/
Time domain
(amplitude over time)

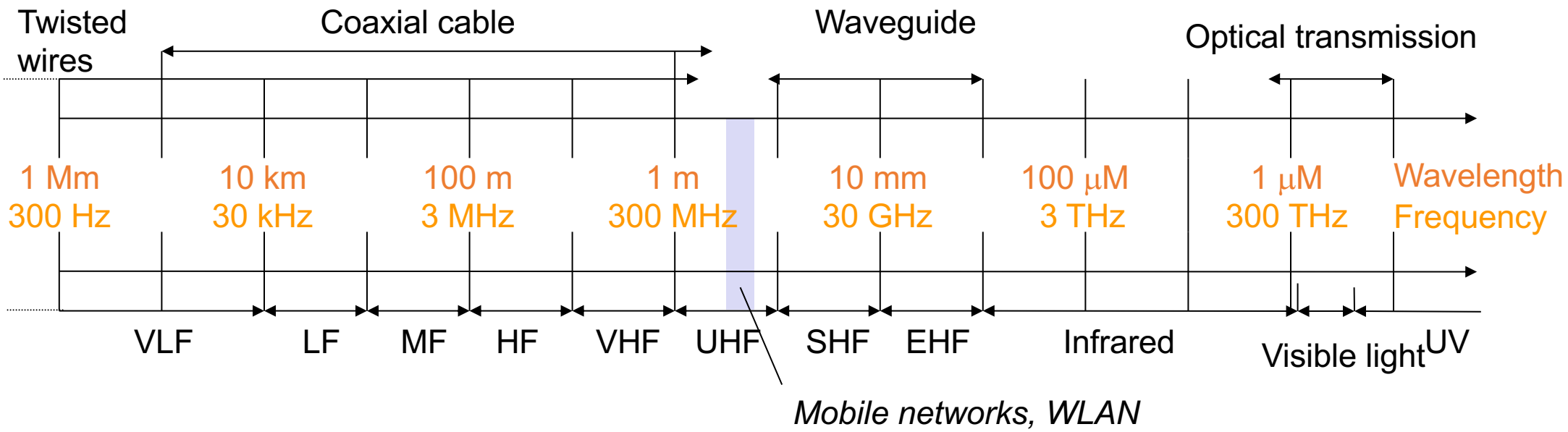
Frequency spectrum/
Frequency domain
(amplitude or phase over frequency)



$I = A \cos \phi$
(In-phase)
$Q = A \sin \phi$
(Quadrature)

Phase state diagram
(Amplitude A and phase ϕ are plotted in polar coordinates)

Frequency ranges (1)



VLF = Very Low Frequency
LF = Low Frequency (long wave radio)
MF = Medium Frequency (medium wave radio)
HF = High Frequency (short wave radio)
VHF = Very High Frequency (VHF radio)

UHF = Ultra High Frequency
SHF = Super High Frequency
EHF = Extra High Frequency
UV = Ultraviolet light

- Relationship between frequency and wavelength

$$\lambda = c/f$$

with wavelength λ,
Speed of light $c \cong 3\text{x}10^8$ m/s,
Frequency f

Frequency ranges (2)

- Antenna size and propagation characteristics (e.g. penetration of materials) depend on wave length / frequency

- VHF/UHF range for mobile radio
  - Manageable, small antennas
  - Propagation conditions favorable for mobile radio (hardly any reflective waves, propagation also possible through walls)

- Frequencies from the SHF range for radio relay links, satellite communication
  - Manageable antenna dimensions with strong bundling effect
  - Larger bandwidths available
  - Penetrability of walls not necessary

- For wireless LANs Frequencies from UHF range to SHF range
  - Also planned up to EHF level
  - Limitation by resonance of molecules (water, oxygen, etc.)
    - strong weather-related damping

Example: Mobile radio frequencies

	Europe	USA	Japan
Cell phones	NMT 453-457 MHz, 463-467 MHz **GSM** 890-915 MHz, 935-960 MHz 1710-1785 MHz, 1805-1880 MHz	**AMPS, TDMA,** CDMA 824-849 MHz, 869-894 MHz **TDMA, CDMA,** GSM 1850-1910 MHz, 1930-1990 MHz	PDC 810-826 MHz, 940-956 MHz, 1429-1465 MHz, 1477 -1513 MHz
Cordless phones	**CT1+** 885-887 MHz, 930-932 MHz **CT2** 864-868 MHz **DECT** 1880-1900 MHz	**PACS** 1850-1910 MHz 1930-1990 MHz **PACS-UB** 1910-1930 MHz	**PHS** 1895-1918 MHz **JCT:** 254-380 MHz
Wireless LANs	**IEEE 802.11** 2400-2483 MHz 5150-5350 MHz, 5470-5725 MHz **HIPERLAN** 5176-5270 MHz	**IEEE 802.**11 2400-2483 MHz, 5150-5350 MHz, 5725-5825 MHz	**IEEE 802.11** 2471-2497 MHz, 5150-5250 MHz

Transmission of digital signals (1)

- Problem: We want to transmit digital signals using sinusoidal analogous signals (radio waves)

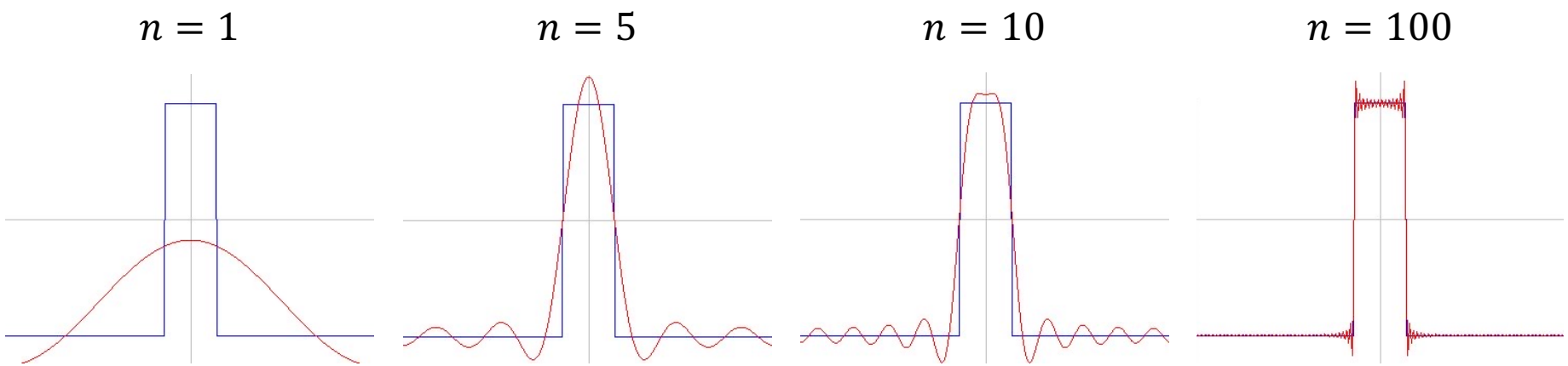
- Naïve solution: A periodic function $x(t)$ can be represented as a sum of sine and cosine functions (Fourier series)

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} [A_n \cos(2\pi n f_0 t) + b_n \sin(2\pi n f_0 t)]$$

  - Base frequency $f_0 = \frac{1}{T}$ is the reciprocal of the period of $x(t)$
  - A_n and B_n are amplitudes of the cosine and sine components of the n-th harmonic

Transmission of digital signals (2)

- Example: Best possible approximation of a square signal

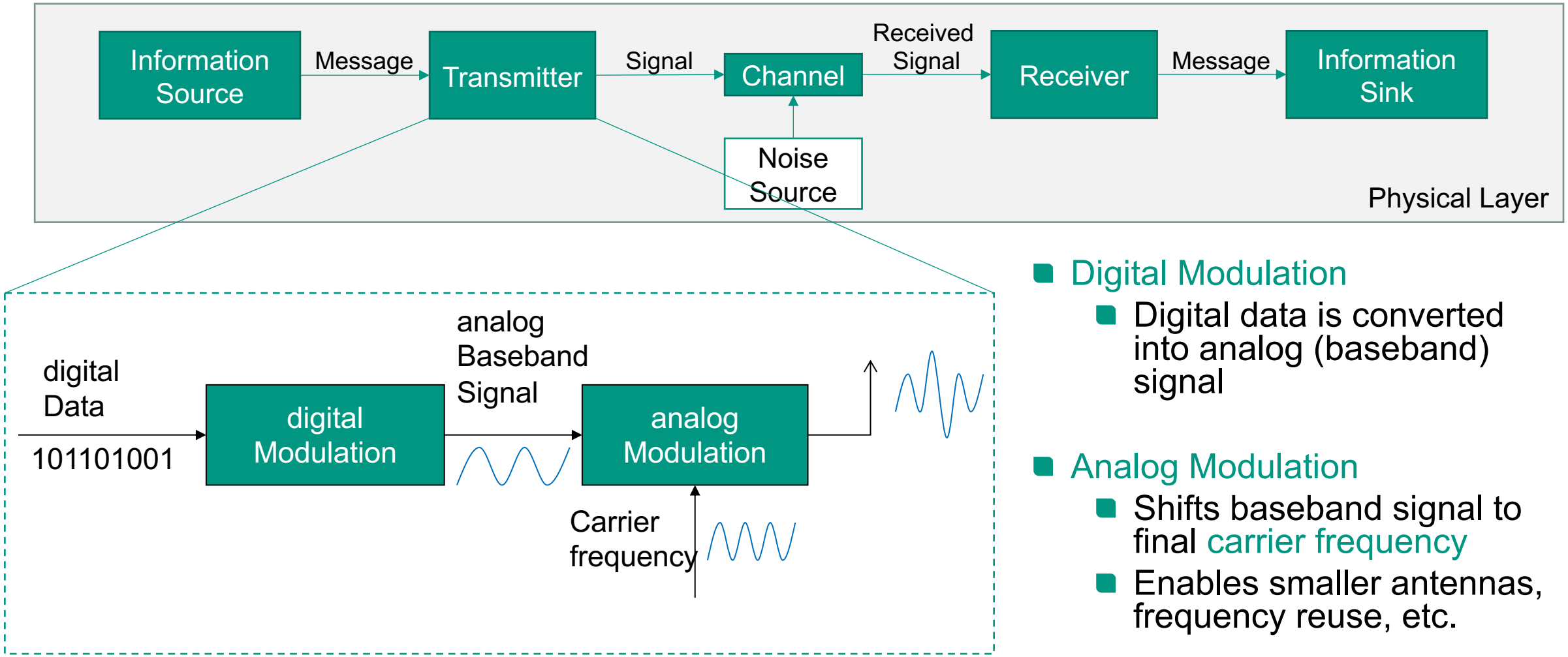


- Observations
  - Better approximation with additional frequency components
  - Sharp signal edges correspond to high-frequency oscillations

- But:
  - Physical medium is bandwidth-limited → Exact representation is not possible
  - A single signal would occupy the entire frequency spectrum
  - Not suitable for practical use!

Solution: Modulation

- Recap Shannon Model



- Digital Modulation
  - Digital data is converted into analog (baseband) signal
- Analog Modulation
  - Shifts baseband signal to final carrier frequency
  - Enables smaller antennas, frequency reuse, etc.

Digital Modulation (Shift Keying)

- A digital signal is prepared for physical transmission by modifying a sinusoidal analog carrier signal $C(t)$.

Amplitude shift keying

Phase shift keying

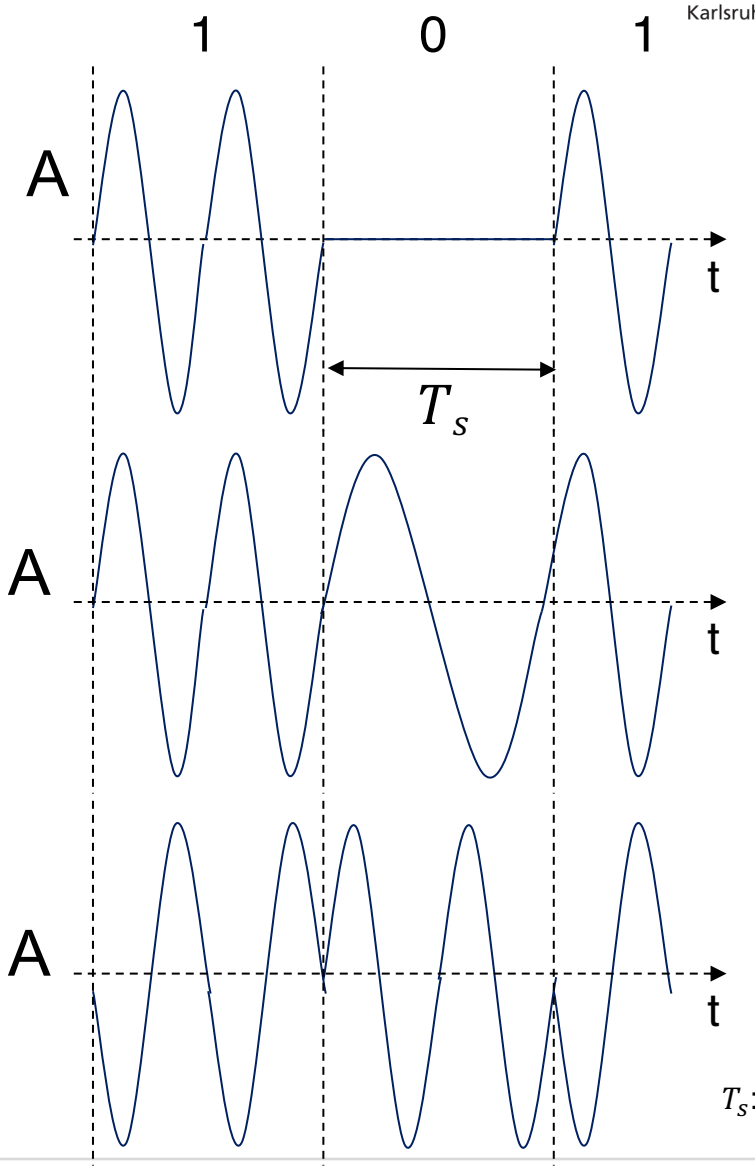
$$C(t) = A\sin(2\pi f t + \phi)$$

Frequency shift keying

- Idea
  - Symbols represent discrete units of user data (e.g., bit or bit groups).
  - Each symbol is mapped to a specific waveform
    - Defines physical signal transmitted over the medium (e.g., specific amplitude, frequency, or phase)
  - The sequence of symbols determines the sequence of waveforms

Digital modulation techniques

- Amplitude shift keying (ASK)
  - Technically simple
  - requires little bandwidth
  - susceptible to faults
- Frequency shift keying (FSK)
  - Requires larger bandwidth (multiple frequencies)
  - Example: Binary FSK (BFSK)
    - Two frequencies 0: f_1 , 1: f_2
- Phase shift keying (PSK)
  - Complex demodulation with carrier recovery
  - Relatively robust



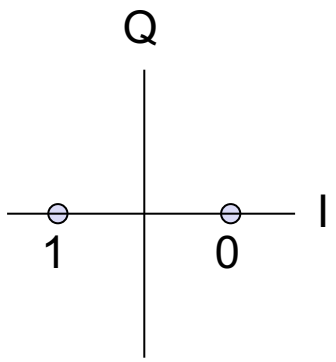
T_s: Symbol Duration

Advanced PSK methods (1)

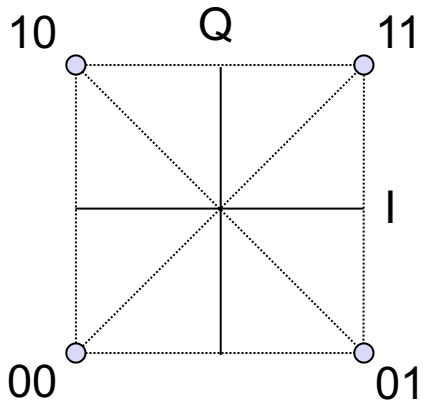
- Binary Phase Shift Keying (BPSK)
  - One date bit is encoded into one signal
    - Bit value 0: Sinusoidal signal
    - Bit value 1: negative sinusoidal signal
  - Simplest phase modulation method
  - Spectrally inefficient
  - Robust, used in satellite systems

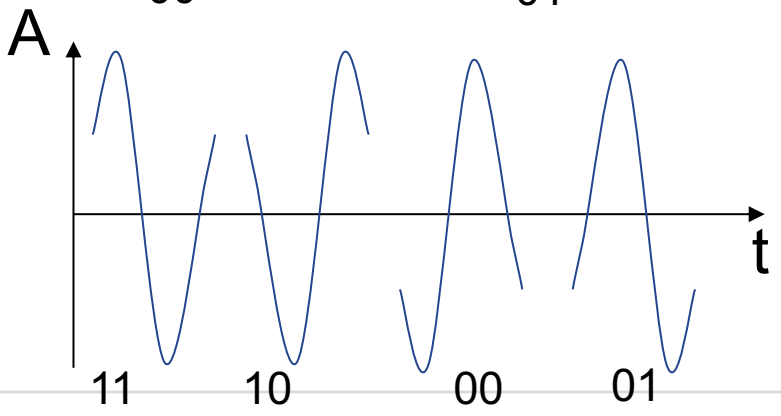
- Quaternary Phase Shift Keying (QPSK)
  - Two data bits are encoded into one symbol
    - Distribution of bits to two channels
    - Separate amplitude modulation on two 90° phase-shifted sinusoidal carriers
      - E.g., First bit cos signal, second bit sin signal
    - Symbol corresponds to sum of both carriers
  - Higher data rates than with BPSK
  - More complex

BPSK:



QPSK:

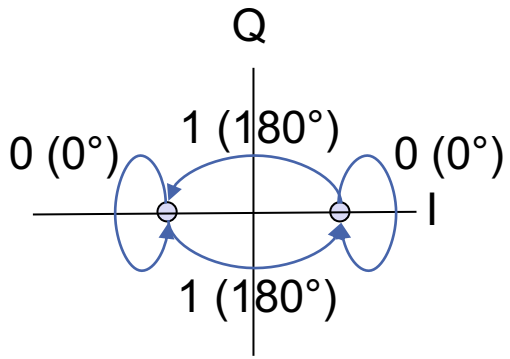




Advanced PSK methods (2)

- Differential Modulation
  - Encode data via phase differences
  - Differential BPSK (DBPSK):
    - 1 bit $\rightarrow \Delta\varphi \in \{0°, 180°\}$
  - Differential QPSK (DQPSK):
    - 2 bits $\rightarrow \Delta\varphi \in \{0°, 90°, 180°, 270°\}$
- Advantages
  - No absolute phase reference needed
  - Robust against phase offsets

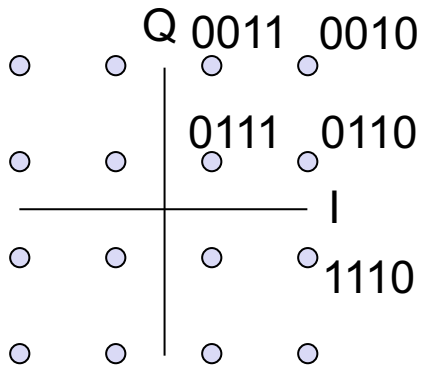
Example DBPSK:



Quadrature Amplitude Modulation (QAM)

- Combination of amplitude and phase modulation by generalization of QPSK
  - Encode n bits per symbol
    - Distribution of bit groups to two channels ($n/2$ bits per channel)
    - Separate amplitude modulation on two 90° phase-shifted sinusoidal carriers, which are then added together (similar to QPSK)
      - $2^{n/2}$ amplitude levels per carrier
      - 2^n discrete states, $n = 2$ corresponds to QPSK

- Example: 16-QAM (4 bits per symbol)
  - The symbols for 0111 and 0010 have have same phase and different amplitude
  - The symbols for 0110 and 1110 have have different phases and same amplitude



Exercise: Modulation

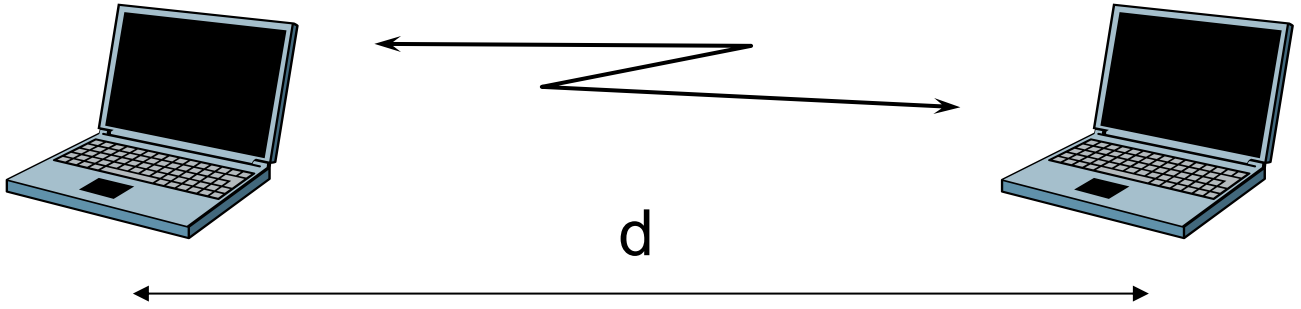
Sketch the resulting signal (amplitude over time) for the transmission of the data bits 010110 using the modulation methods ASK, FSK, BPSK, and QPSK.

What are the advantages and disadvantages of these methods?

Signal Propagation

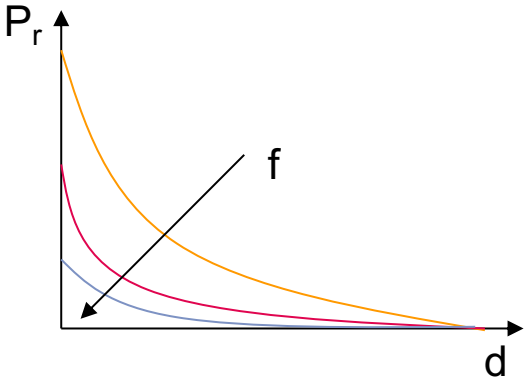
Signal propagation

- Signal propagation in ideal case
  - Not influenced by any obstacles



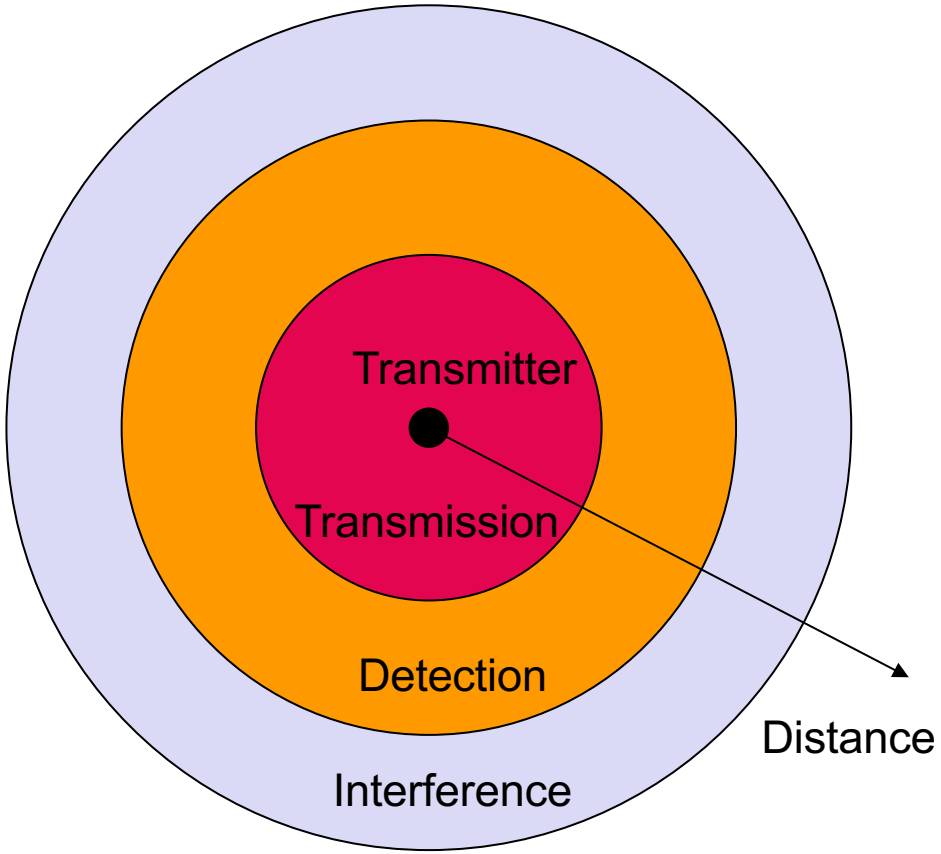
$$P_r \sim \frac{1}{d^2} P_t$$

  - Received power decreases with $1/d^2$
    - d = distance between transmitter and receiver
    - P_t = transmitted signal power (or transmit power)
    - P_r = received signal power (or received power)
  - Proportionality factor depends on frequency



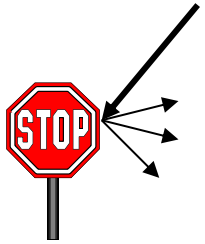
Signal propagation

- Different areas of signal propagation
  - Transmission range
    - Communication possible
    - Low error rate
  - Detection range
    - Signal detection possible
    - No communication possible
  - Interference range
    - Signal cannot be detected
    - Signal adds to background noise
- Idealized representation!
  - Assumes obstacle-free propagation
- Areas are…
  - …not sharply separated from each other!
  - …not necessarily circular!



Signal propagation - Influencing factors

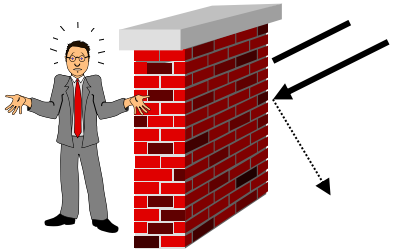
- Scattering at small obstacles
  - Splitting into several weaker signals



- Diffraction
  - Wave bent into shadowing area of an obstacle
  - Diffraction stronger with larger ratio of wavelength to dimensions of the obstacle
    - Negligible above about 5 GHz

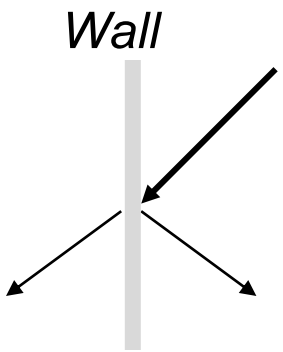
Influencing factors

- Shadowing due to obstacles
  - Extreme form of damping



Schadowing
(idealized)

- Reflection
  - Attenuation of the signal

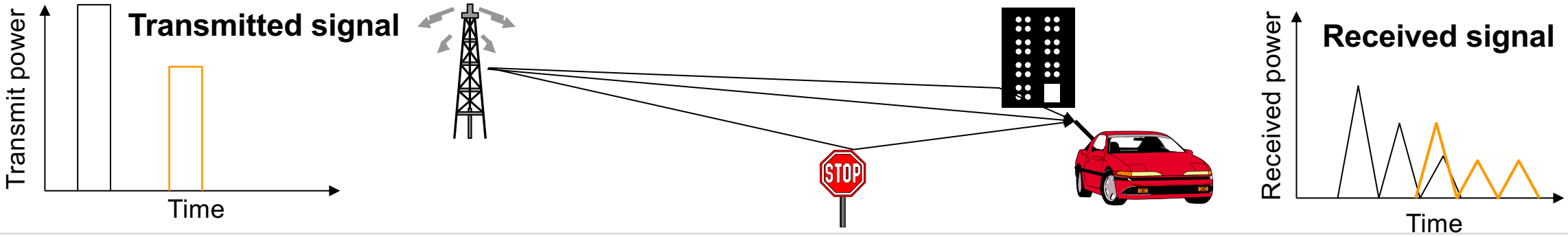


Reflection
(idealized)

- Clearance damping
  - Atmosphere changes its properties due to weather conditions
    - Highly frequency-dependent
      - E.g. from about 12 GHz in fog or rain, strong attenuation due to scattering and absorption of electromagnetic waves by water droplets

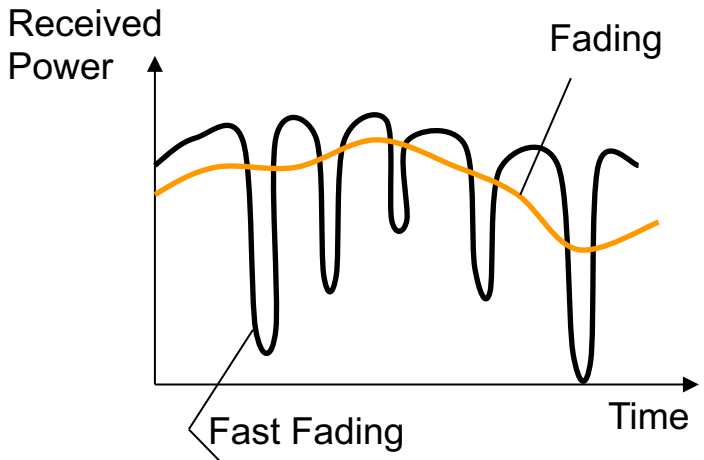
Influencing factors

- Multipath propagation
  - Due to reflection, scattering and diffraction, signal reaches receiver on several paths (of different lengths)
    - Delay spread: Difference between arrival time of first and last signal component
  - Direct and phase-shifted signal components are received (see below)
    - Attenuated signal depending on phase position
  - Signal is dispersed over time (time dispersion)
    - Interference between neighboring symbols (inter symbol interference, ISI)



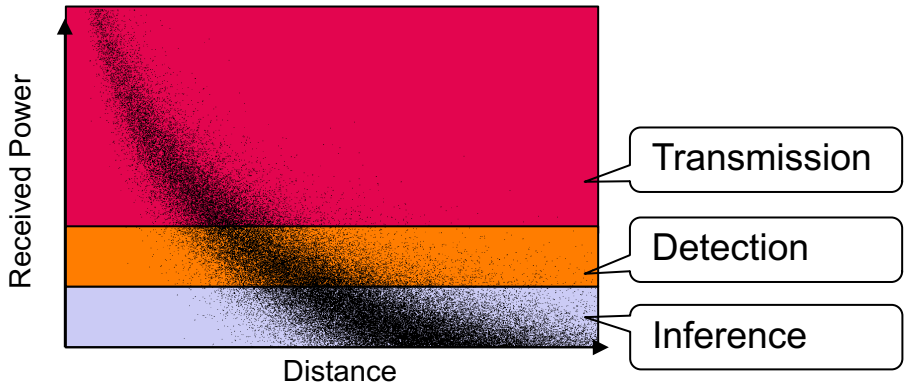
Influencing factors

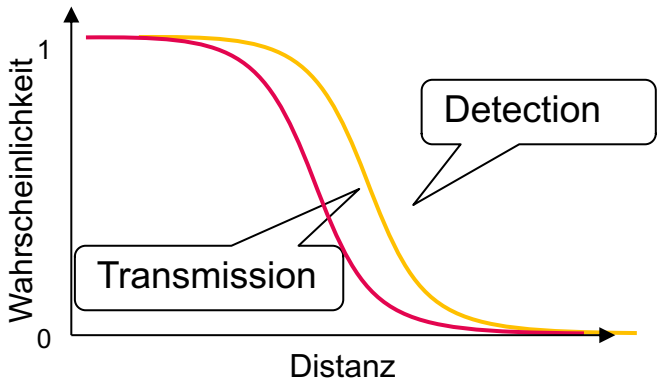
- Fading
  - Fluctuations in receive power caused by propagation effects
  - Transmission quality changes:
    - With mobile station position
    - Over time
  - Changing transmission paths
    - Different distance, obstacles, propagation paths, delay spreads, phase shifts of signal components, …
- Fast Fading
  - Rapid fluctuations in received signal power
  - Caused by destructive interference of multiple signal paths
  - Additionally influenced by:
    - Distance to base station
    - Dynamic (distant) obstacles
- Slow Fading
  - Slow variations in average received signal power
  - Caused by large-scale environmental changes (e.g., distance, buildings, terrain)



Refinement of the signal propagation model

- Numerous factors influencing the signal (see above)
  - Dispersion-related interference
  - Signal level can hardly be predicted exactly
  - Signal level can be better modeled by distribution
- Consequence for transmission between two stations
  - Transfers may fail
  - Transmission may only be unidirectional
  - Average reception probability decreases with distance

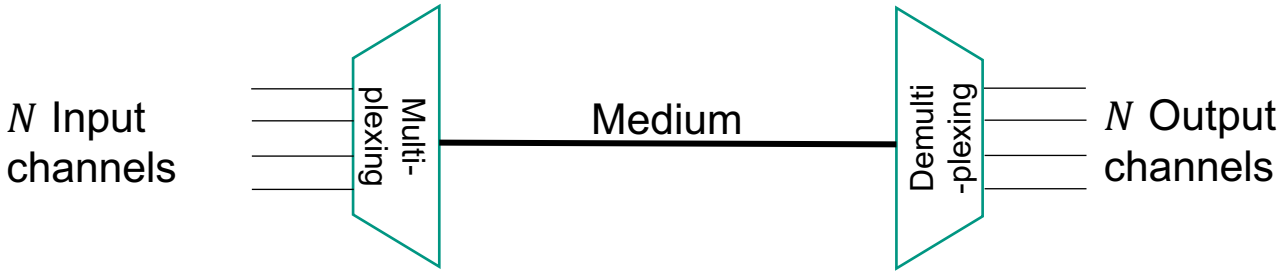




Multiplexing

Multiplexing (1)

- Definition: Multiplexing is a technique to combine multiple channels (signals or data streams) for transmission over a single medium, allowing them to be transmitted simultaneously and efficiently



Multiplexing (2)

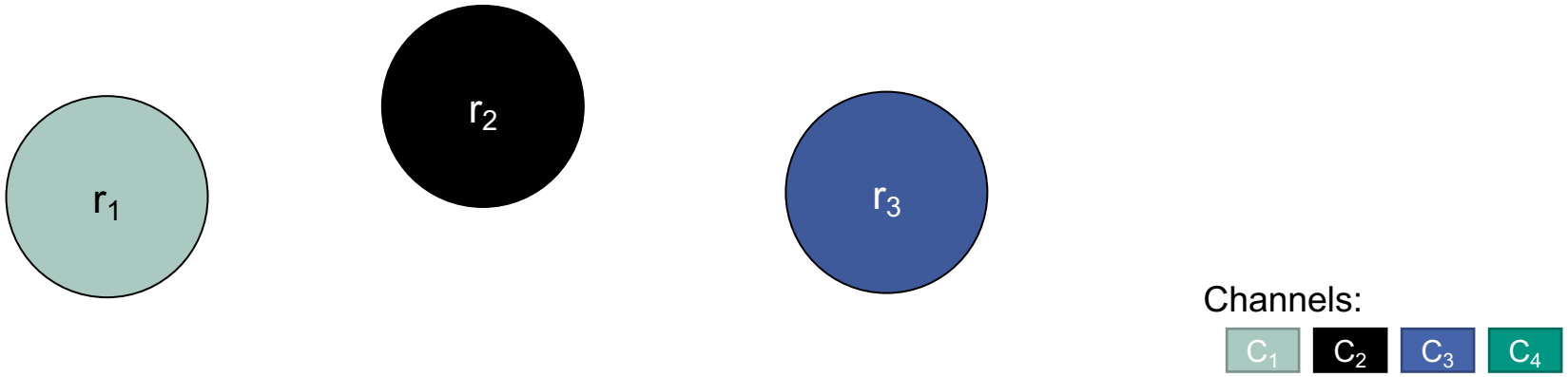
- Multiplexing is possible in four dimensions:
  - Space (r) – Space Division Multiplexing (SDM)
  - Time (t) – Time Division Multiplexing (TDM)
  - Frequency (f) – Frequency Division Multiplexing (FDM)
  - Code (c) – Code Division Multiplexing (CDM)

- For channels belonging to different senders (users, devices) we use the term Multiple Access instead of multiplexing
  - E.g., Time Division Multiple Access (TDMA)

Space Division Multiple Access (SDMA)

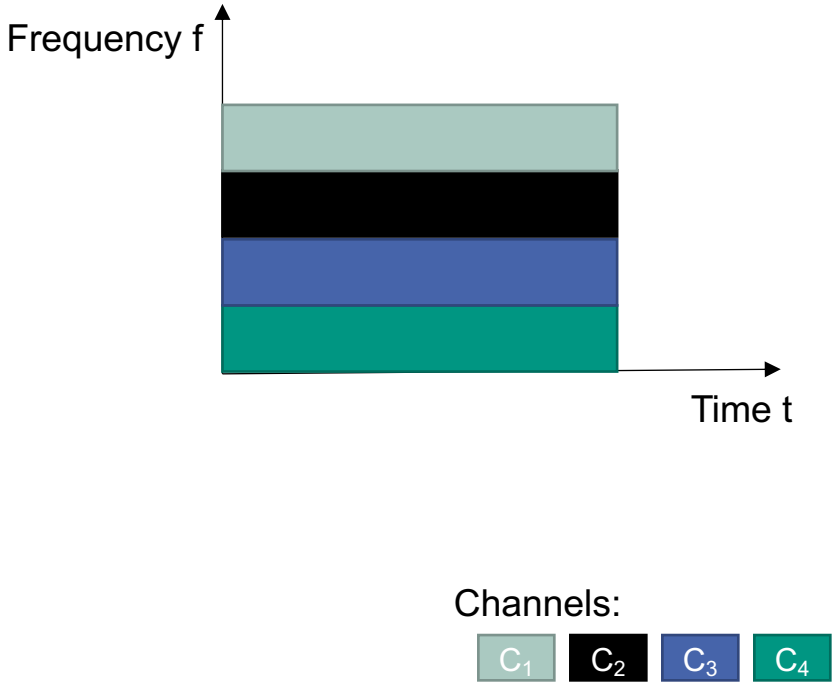
- Approach
  - Exploitation of limited transmission / interference range of wireless transmissions
  - All channels use the same frequency at the same time at different places with distance larger than interference range

- Example:
  - Reuse of radio frequencies, cell structure in cellular networks



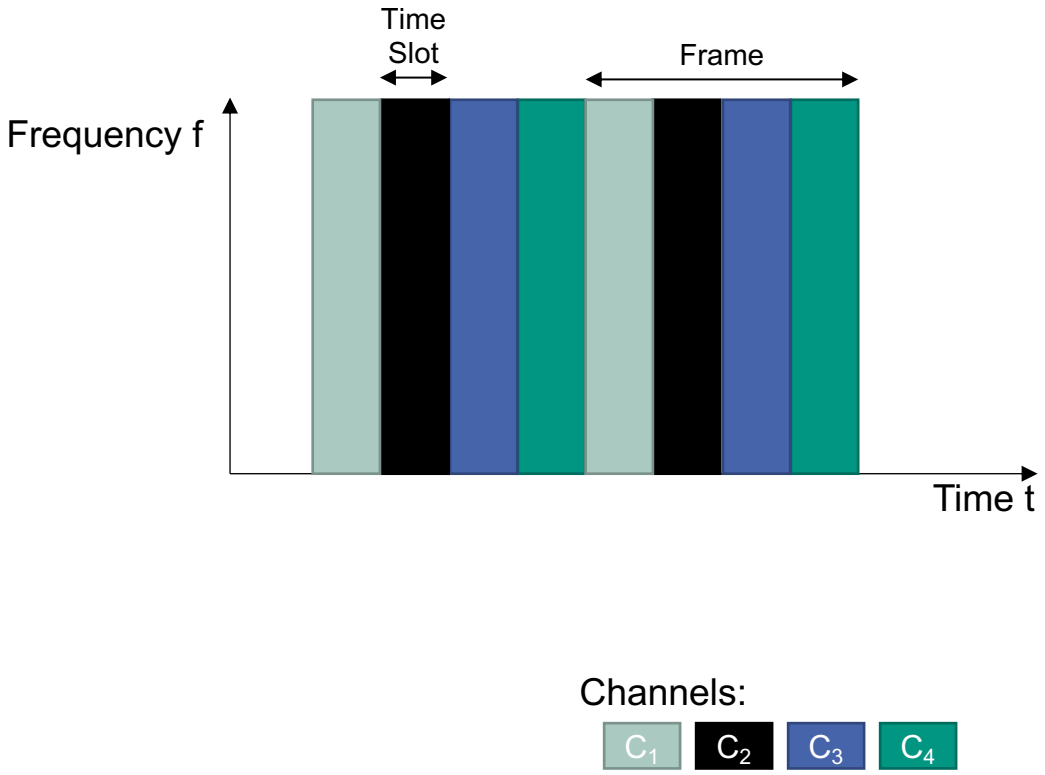
Frequency Division Multiple Access (FDMA)

- Approach
  - Available bandwidth is divided into multiple frequency bands (or carriers or subcarriers)
  - Each channel uses one frequency band for entire time
- Example:
  - Frequencies in radio and TV, WLAN channels



Time Division Multiple Access (TDMA)

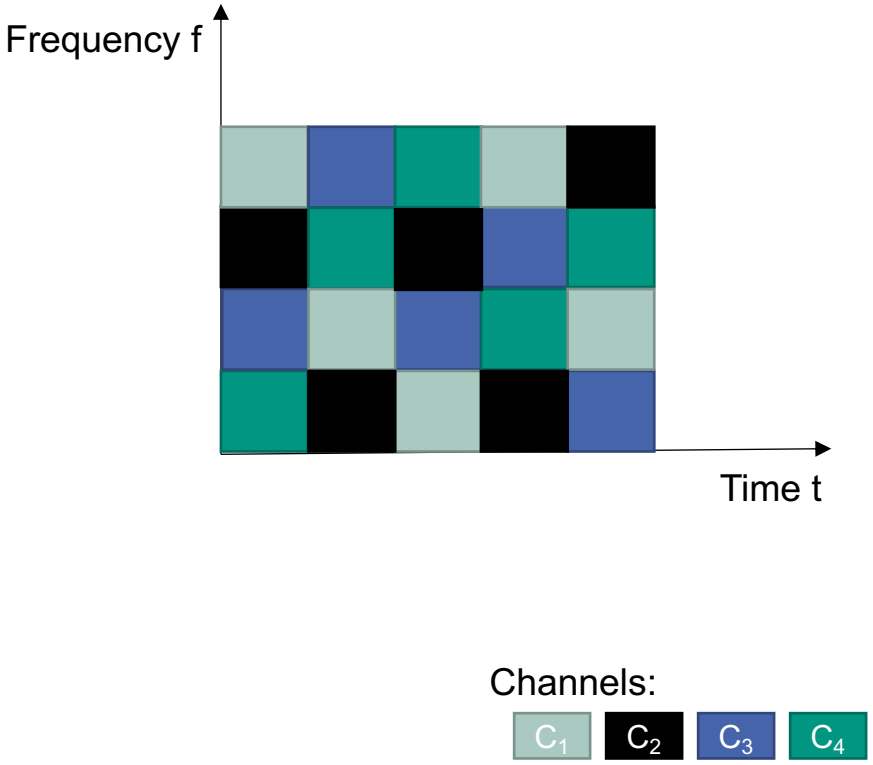
- Approach
  - Time is divided into time slots
    - Often grouped into repeating frames
  - Channels use medium in different time slot
  - Length of time slots can be identical or different
  - Time slots assignment can be controlled or contention based
- Example:
  - WLAN, voice channel in GSM



Time and Frequency Division Multiple Access

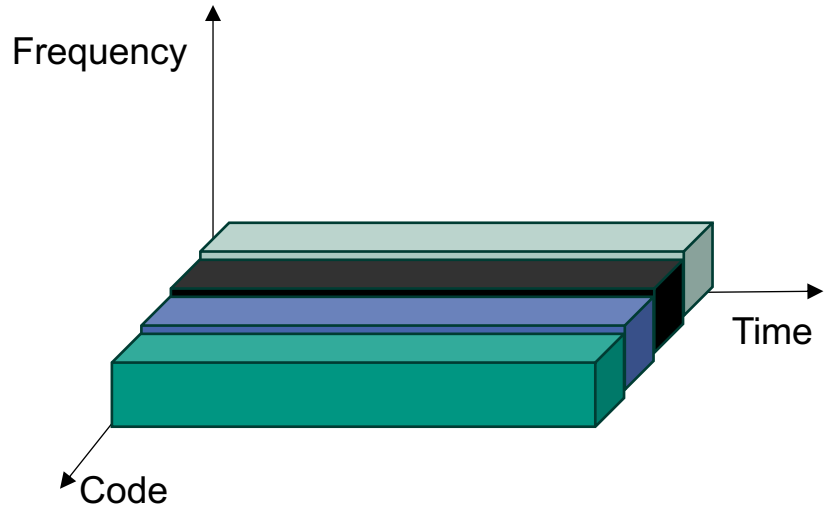
- Approach
  - Combination of FDMA and TDMA
  - Assignment of a frequency band per time slot for a channel

- Example:
  - Bluetooth



Code Division Multiple Access (CDMA)

- Approach
  - All channels use the same frequency at the same time
  - Each channel uses unique pseudo-random bit sequence (code, chipping sequence)
    - Bits of the code sequence are referred to as chips
  - Data bits are multiplied with each chip, result is transmitted
    - Transmission of single data bit requires transmission of multiple encoded bits
    - Length of code is called spreading factor
  - Receiver can use code and correlation function to restore original data bits



- Example
  - UMTS, early WLAN standards

Channels:

CDMA - Example

- Sender A
  - Sends $A_d = 1$, code $A_k = 010011$ ("0"= +1, "1"= -1)
  - Transmit signal $A_s = A_d \bullet A_k = -1 \bullet (+1, -1, +1, +1, -1, -1) = (-1, +1, -1, -1, +1, +1)$
- Sender B
  - Sends $B_d = 0$, code $B_k = 110101$ ("0"= +1, "1"= -1)
  - Transmit signal $B_s = B_d \bullet B_k = 1 \bullet (-1, -1, +1, -1, +1, -1) = (-1, -1, +1, -1, +1, -1)$
- Both signals overlap (add up) during transmission
  - Interference and noise not considered here
  - Both signals have the same signal strength
  - $A_s + B_s = (-2, 0, 0, -2, +2, 0)$

[2.1]

CDMA - Example

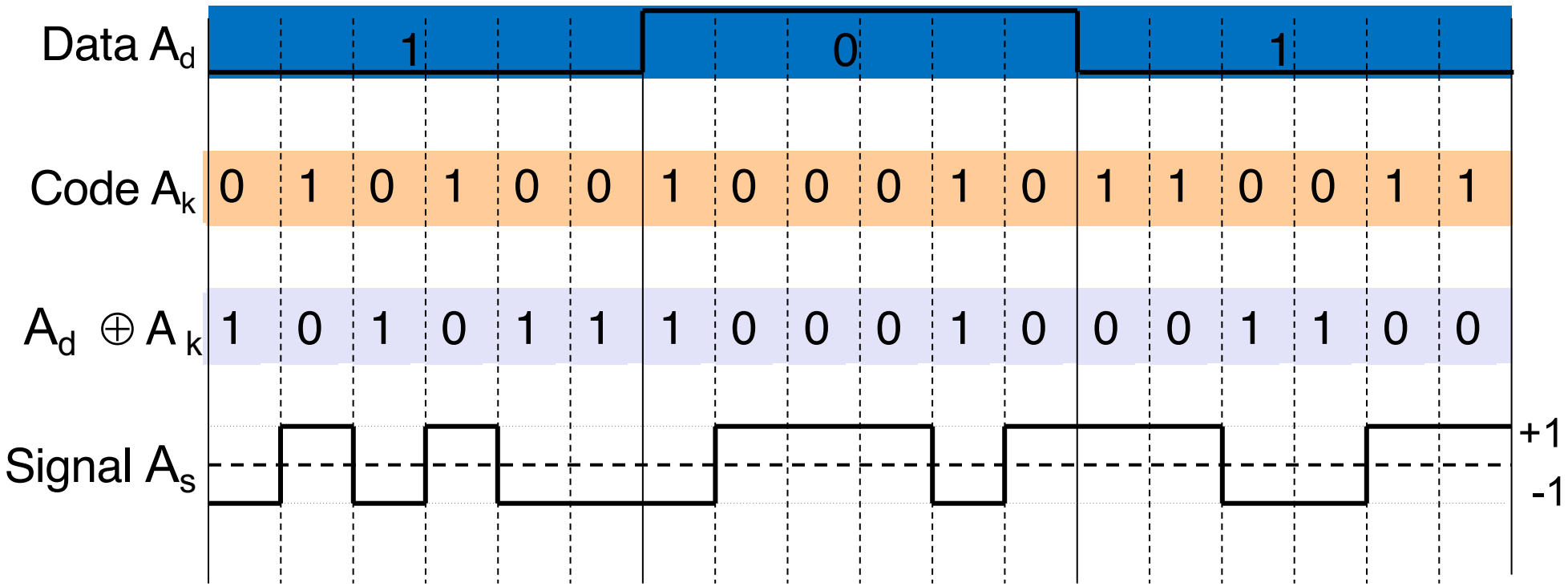
- Receiver wants to receive Transmission A
  - Applies key A_k bit by bit (inner product)
  - $A_e = (A_s + B_s) \bullet A_k = (-2, 0, 0, -2, +2, 0) \bullet (+1, -1, +1, +1, -1, -1) =$
    $- 2 + 0 + 0 - 2 - 2 + 0 = - 6$
  - Result is clearly less than 0, therefore the bit sent was a "1"

- Same holds for Transmission B
  - $B_e = (-2, 0, 0, -2, +2, 0) \bullet B_k = 2 + 0 + 0 + 2 + 2 + 2 + 0 = 6$, i.e. "0"

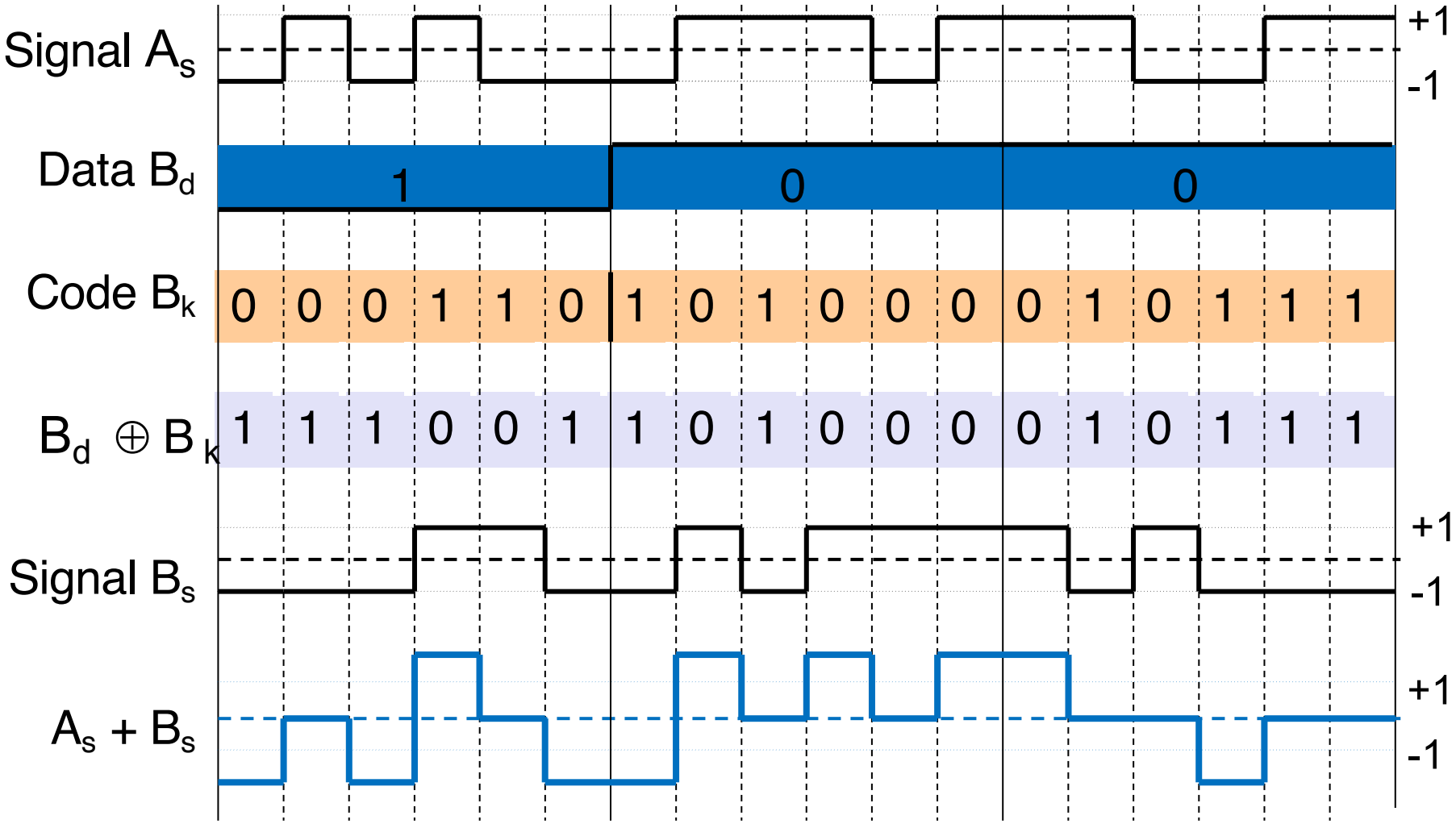
[2.1]

CDMA – Example with longer Codes

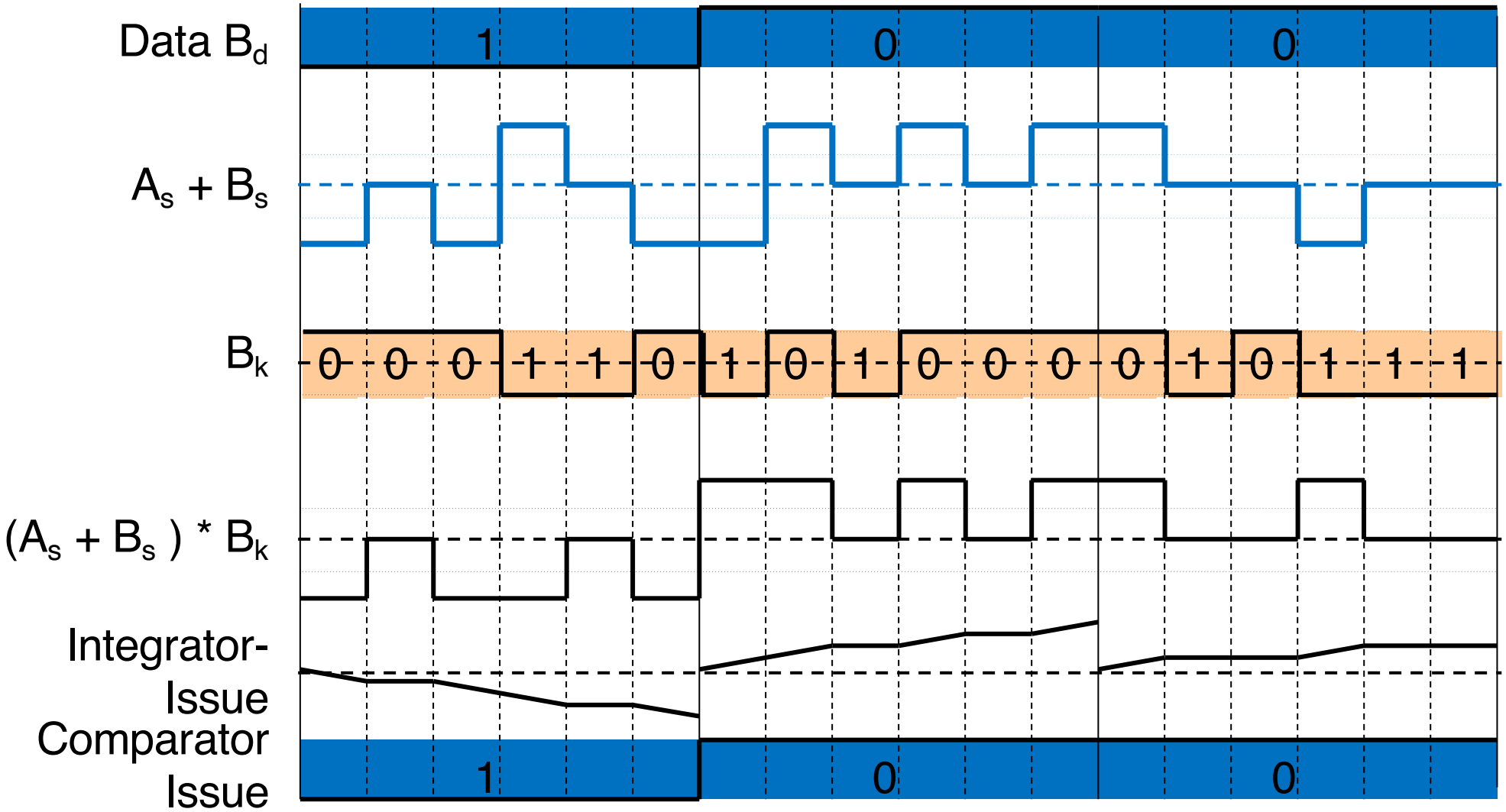
- Signal level in the example: 0 - positive level (+1); 1 - negative level (-1)



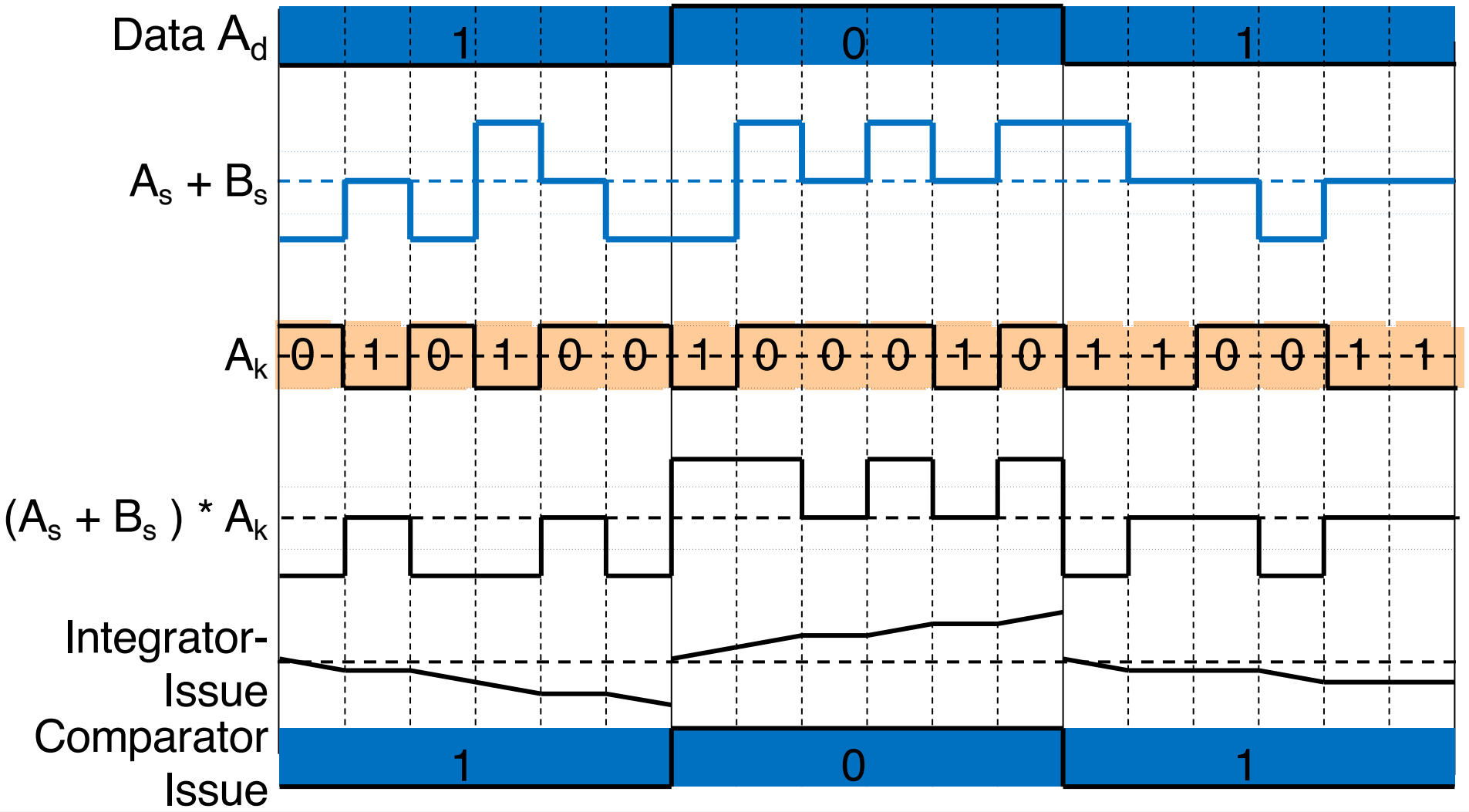
CDMA - Example



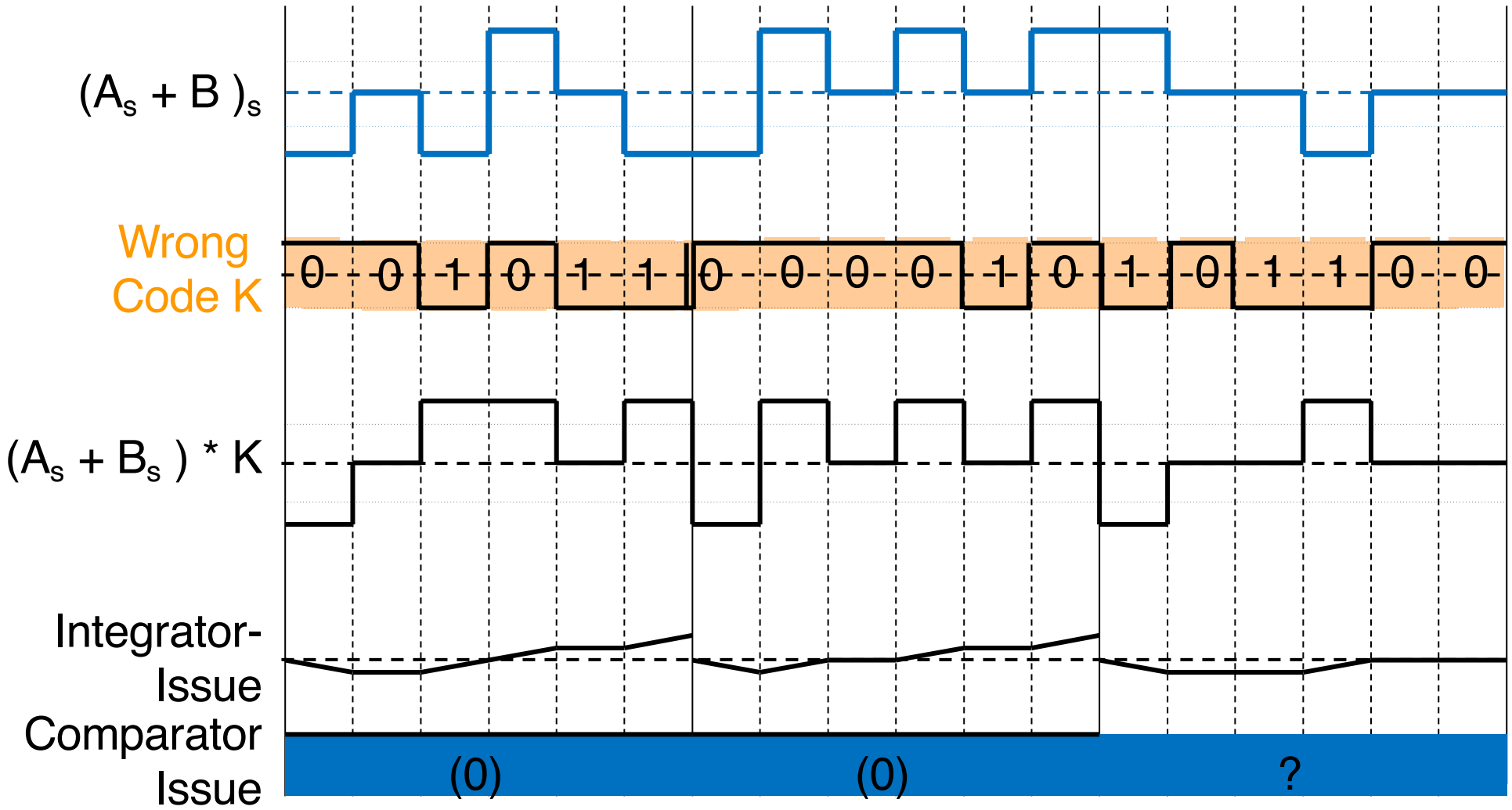
Receiver reconstructs data from B



Receiver reconstructs data from A



Receiver uses wrong code



Exercise: Code Multiplexing

Sender A uses the code $A_k = (1, -1, -1, 1)$, and sender B uses the code $B_k = (1, 1, -1, -1)$. The transmitted data bits are $A_d = (1, -1)$ and $B_d = (-1, -1)$. Here, assume that the data bit 0 corresponds to a level of $+1$, and data bit 1 corresponds to a level of -1.

- What do the transmitted signals of the two stations look like individually and in sum?
- What value appears in the integrator of a receiver that uses A_k after receiving each of the two data bits?

Orthogonal Codes

- Does CDMA work for any combination of codes?
  - No – we need Orthogonal Codes
    - Code sequences with zero cross-correlation
    - Inner product is zero

- Example:

$$A_k = (+1,\ -1,\ +1,\ +1,\ -1,\ -1)$$
$$B_k = (\ -1,\ -1,\ +1,\ -1,\ +1,\ -1)$$
$$A_k \bullet B_k = \ -1\ \ +1\ \ +1\ \ -1\ \ -1\ \ +1 = 0$$

- Signals encoded with different orthogonal codes do not interfere with each other

How to find orthogonal codes?

- Walsh-Hadamard matrix recursively defined by

$$C_1 = [1] \qquad C_{2N} = \begin{bmatrix} C_N & C_N \\ C_N & -C_N \end{bmatrix}$$

- Example:

$$C_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad C_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

- Row of the matrix is called Walsh Sequence
  - Pairwise orthogonal

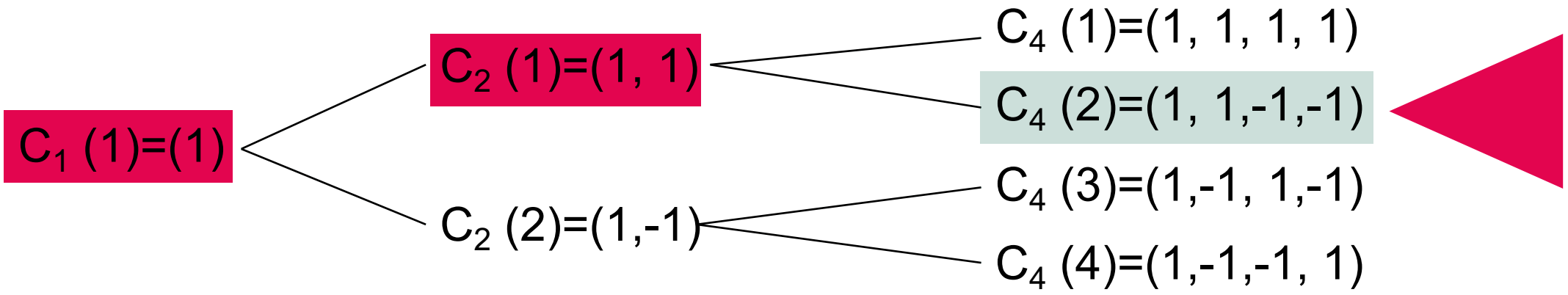
[2.4]

Orthogonal Variable Spreading Factor (OVSF)

[2.4]

- Motivation
  - Different spreading factors for different channels → Different data rates
- Representation of the codes as a tree
  - "Upper" child: Parent sequence concatenated 2x
  - "Lower" child: parent sequence concatenated with its inverse

  - Predecessors and successors may not be used for each selected sequence



Example Application OVSF

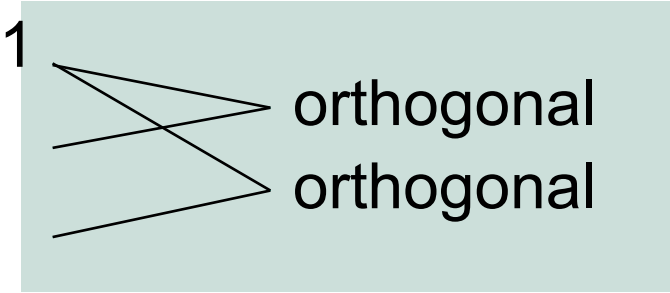
- Valid selection: C_2 (1)=(1,1), C_4 (3)=(1,-1,1,-1), C_4 (4)=(1,-1,-1,1)

[2.4]

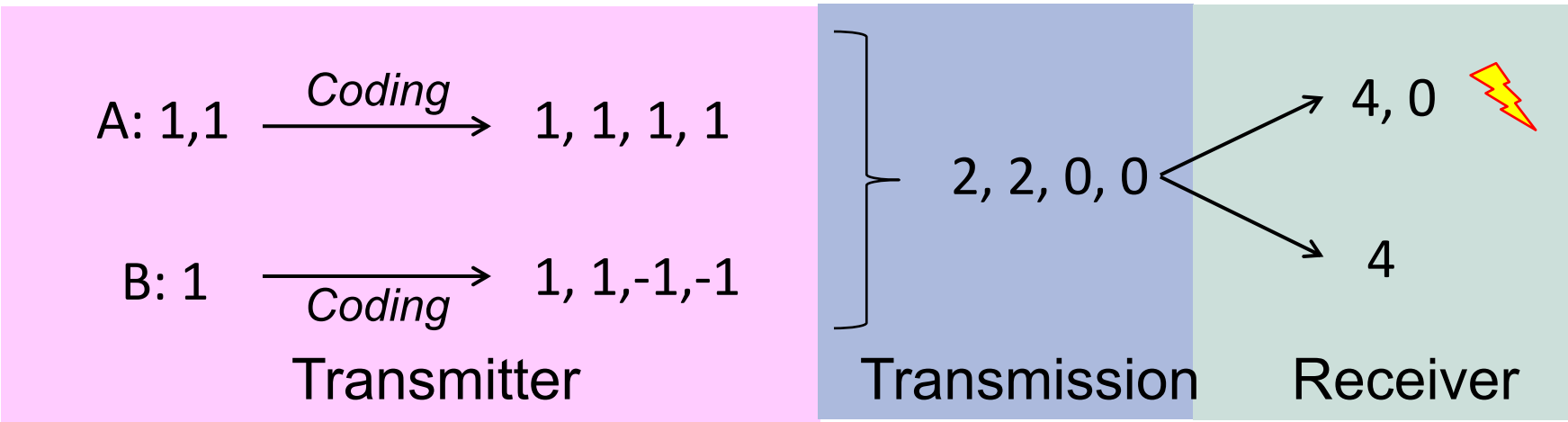
A, selected sequence, repeats 1, 1 |1, 1

B, selected sequence 1,-1, 1,-1

C, selected sequence 1,-1,-1, 1



- Invalid selection: C_2 (1)=(1,1), C_4 (2)=(1,1,-1,-1) *(all bits in signal representation)*



Exercise: Orthogonal Codes

If we need codes of length 4 for one group of transmitters and codes of length 8 for another group, and all codes must be perfectly orthogonal, how many stations of each group can be served at most when using the OVSF method?

Formulate an equation that expresses this relationship!

Pseudo-Orthogonal Codes

- In practice, often strict orthogonality cannot be maintained
  - e.g., due to timing offsets or limited code availability

- Solution: Pseudo-orthogonal codes
  - Low – but non-zero – cross-correlation
  - May cause some interference, but still enable simultaneous transmissions with manageable signal separation

- Example:
  - Gold Codes in UMTS

Comparison SDMA/TDMA/FDMA/CDMA

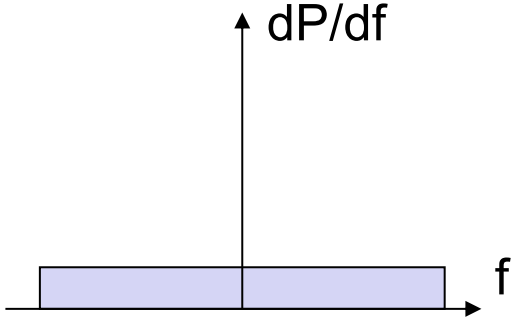
Approach	SDMA	TDMA	FDMA	CDMA
Idea	Division of the room into cells/sectors	Division of the transmission times into disjoint slots, request-controlled or fixed	Division of the frequency range into disjoint bands	Band spreading through individual codes
Participants	Only one participant can be continuously active in a sector	Participants are active one after the other for a short time	Each subscriber has a frequency band, uninterrupted	All participants can be active at the same time in the same place without interruption
Signal isolation	Cell structure, directional antennas	In the time domain through synchronization	In the frequency range through filters	Code plus special receiver
Advantages	Very simple in terms of planning, technology, capacity increase	Established, fully digital, versatile	Simple, established, robust, plannable	Flexible, requires less frequency planning, smooth handover
Disadvantages	Inflexible, typically fixed antennas	Protection times necessary due to multipath propagation, synchronization	Low flexibility, frequencies in short supply	Complex receiver, requires precise control of the transmission power
Remark	Only useful in combination with TDMA, FDMA or CDMA	Standard in fixed networks, often combined with FDMA in mobile networks	Combined with TDMA in e.g. GSM	Used in UMTS

Spread Spectrum Techniques

Types of interference

Wideband interference

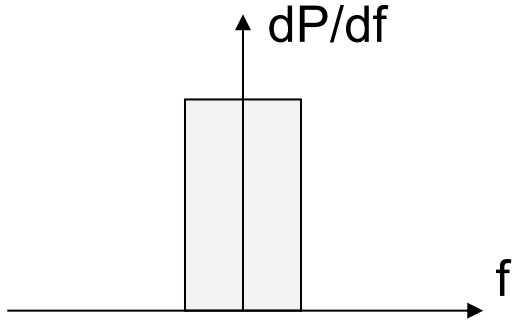
- Constant power density across all frequencies



- Caused by: Motors, thyristor circuits, inverters, computers, contacts of electrical switches, voltage regulators, pulse generators, thermostats, welding equipment

Narrowband interference

- Selective power density on individual frequencies



- Caused by: Uninterruptible power supplies, mains hum, ballasts, test equipment, cellular networks, Radio/TV transmitters, signal generators, microwave devices, ...

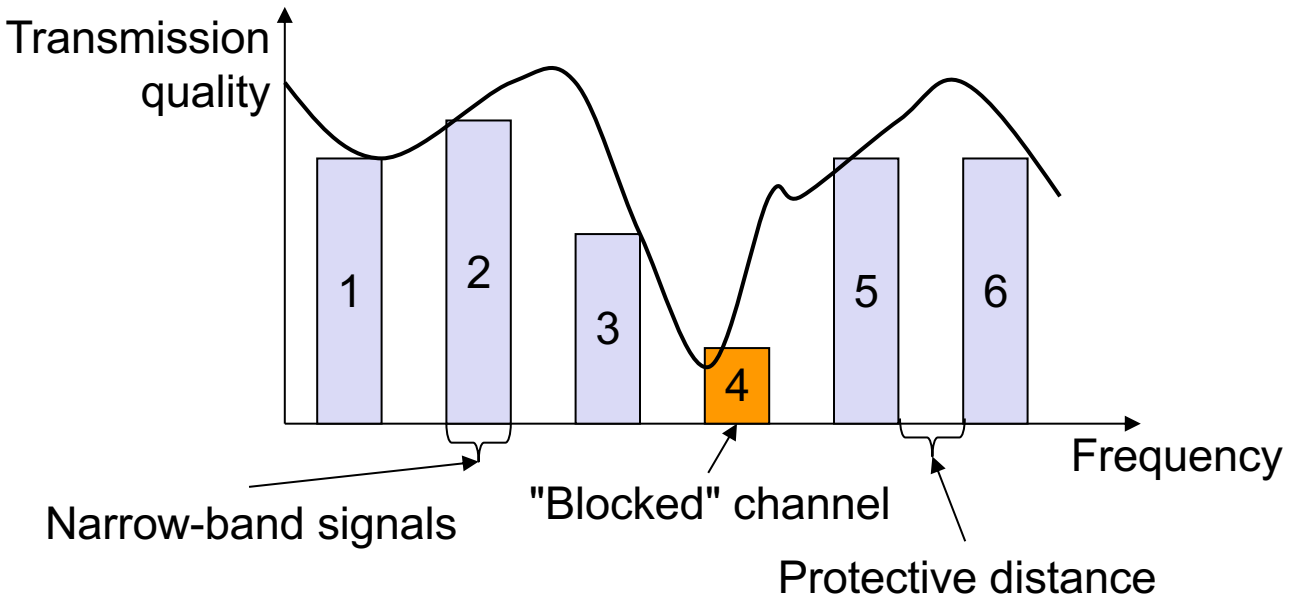
Broadband interference

narrowband interference

[2.7]

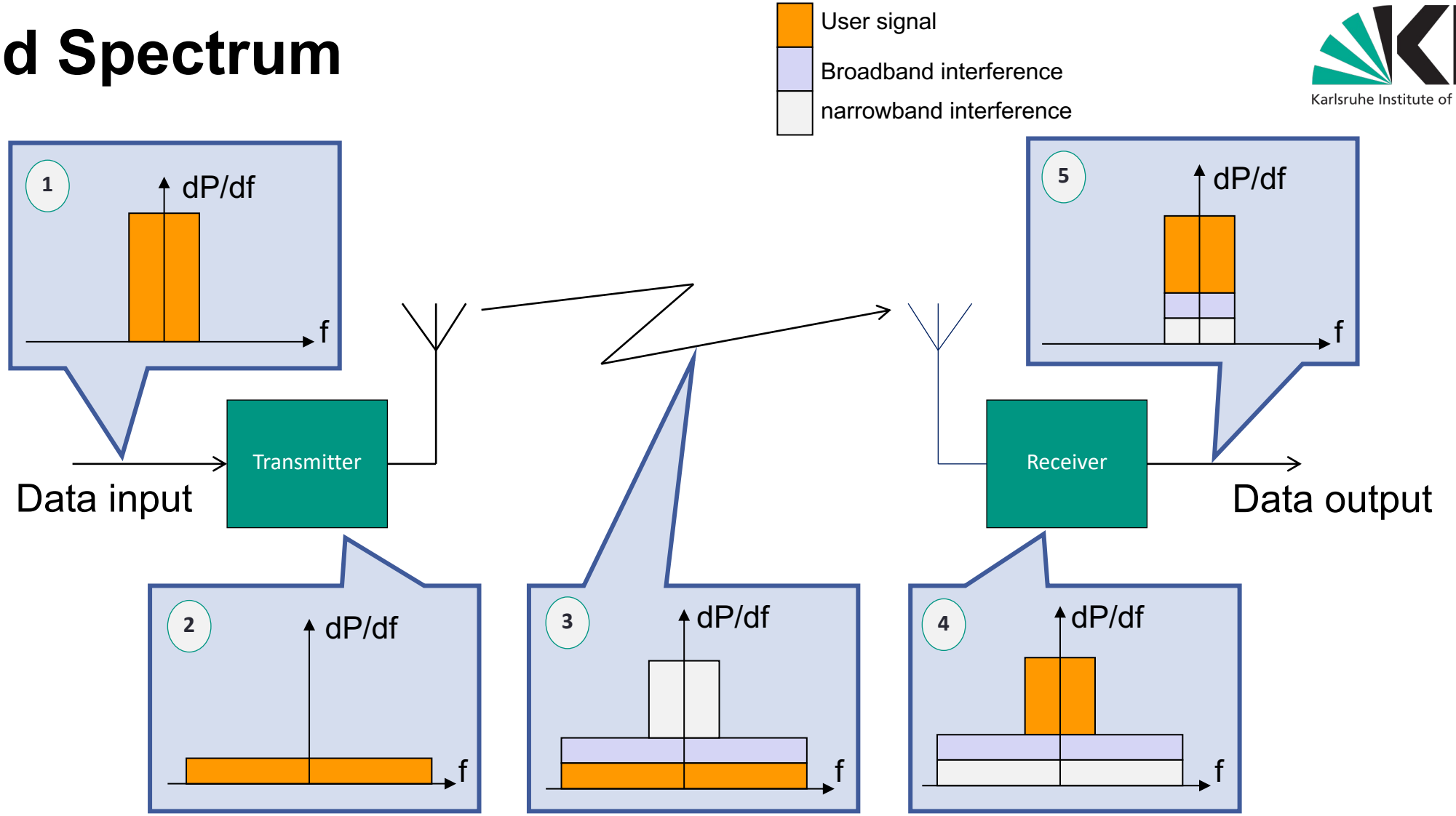
Effects of interference

- Reduction in transmission quality
  - Narrowband channels can be completely blocked



- Solution: "Spread" the signal over a wide frequency band!

Spread Spectrum



(1) Narrowband signal for transmission
(2) Spread Spectrum
(3)Superposition due to interference

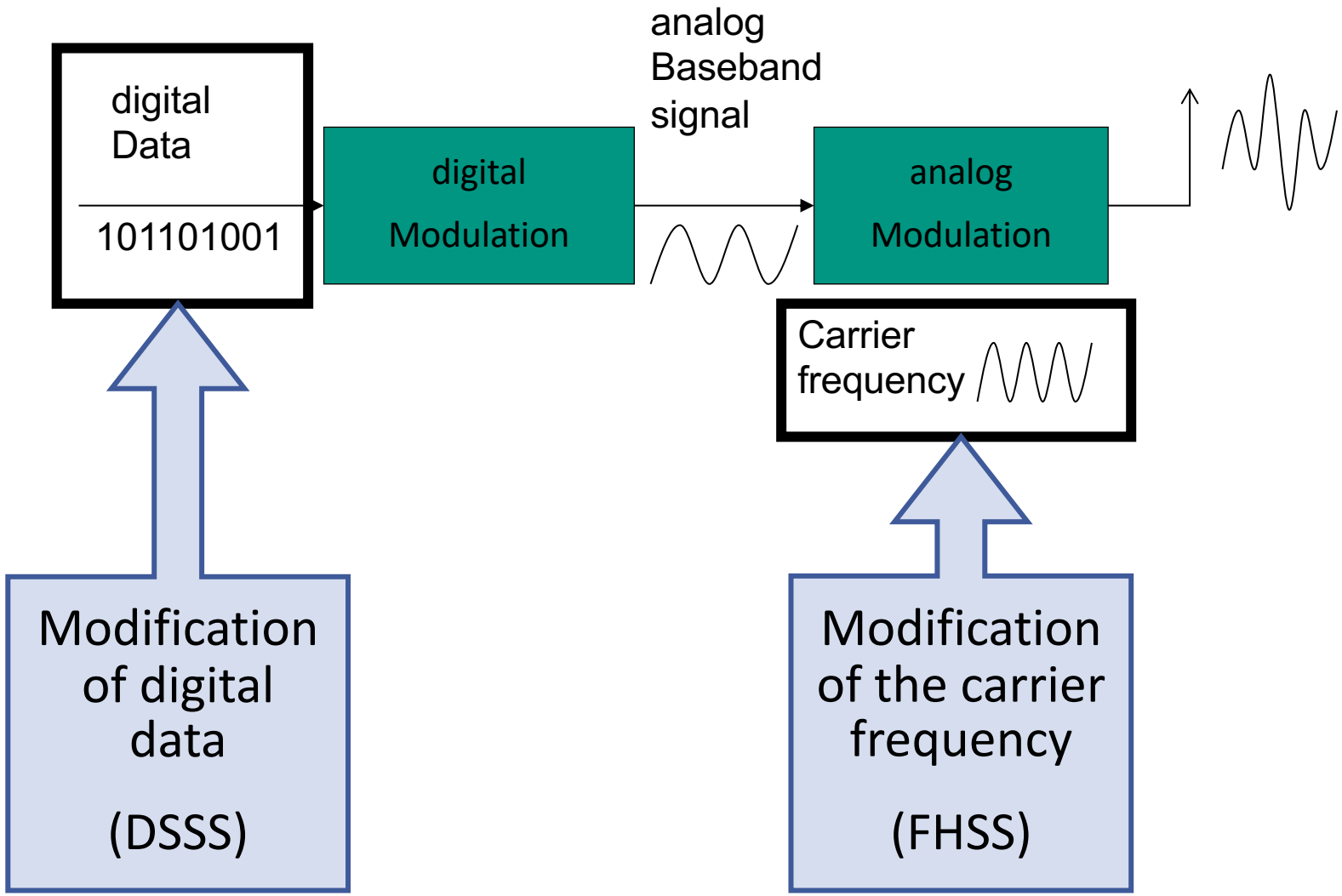
(4)Spreading the signal
(5)Bandpass

[2.10]

Spread spectrum - Advantages

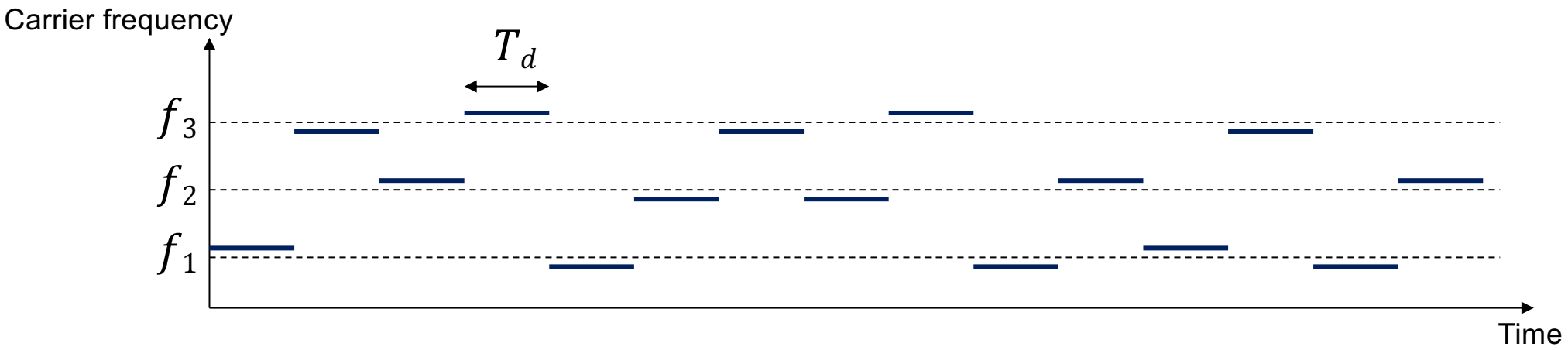
- Insensitivity to
  - Different types of interference
  - Multipath propagation due to reflections
- Coexistence of many users with little mutual interference
- Hiding and encrypting signals

Spread spectrum - How to do it?



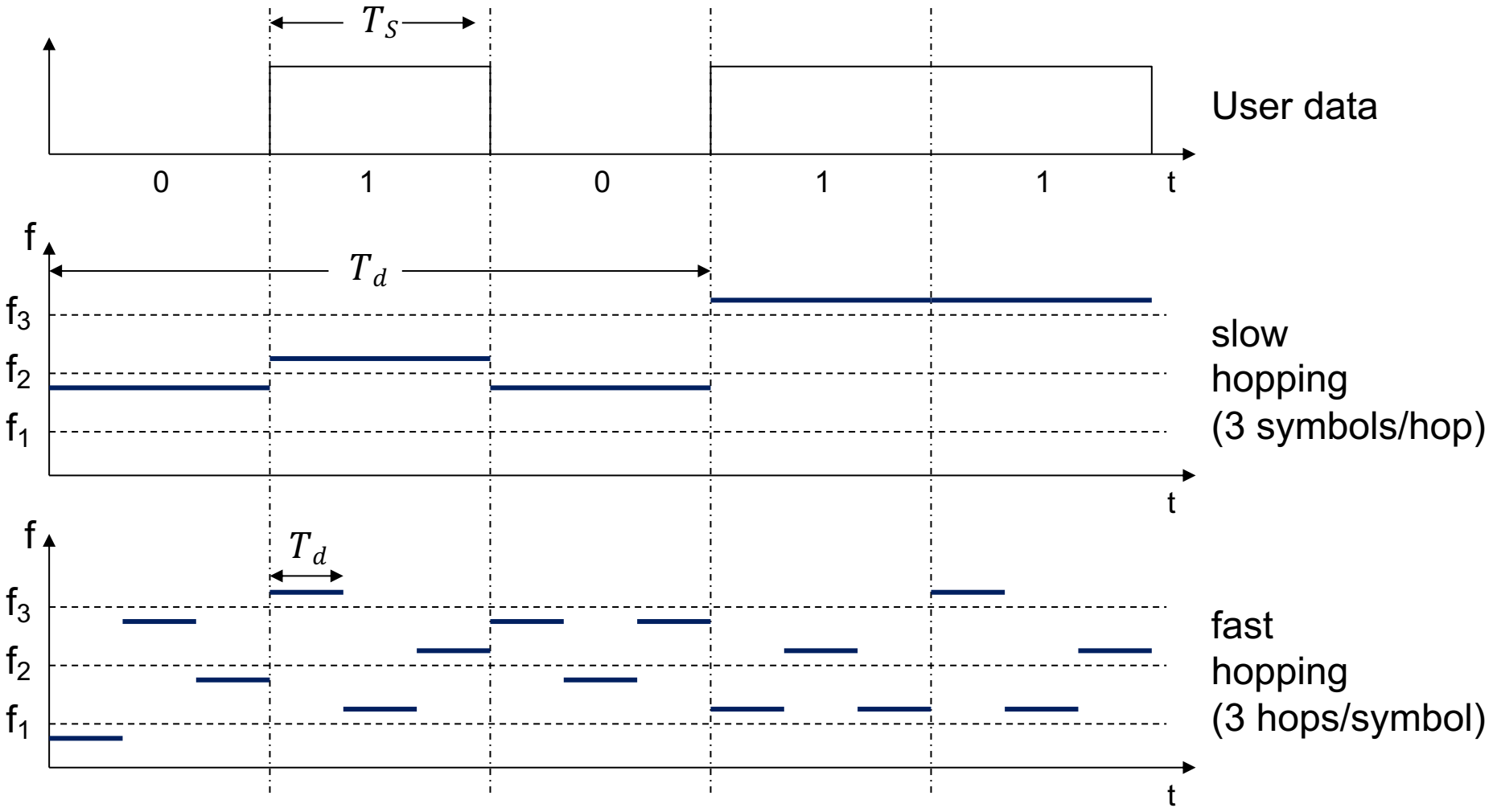
Frequency Hopping Spread Spectrum (FHSS)

- Application of frequency multiplexing: Spreading by changing between multiple carrier frequency
  - Hopping sequence: Pattern of frequency changes
  - Often determined by pseudo-random number sequence, known by sender and receiver
  - Dwell time T_d : Time on a carrier frequency
- Example of a hopping sequence over three carrier frequencies f_1, f_2, f_3:



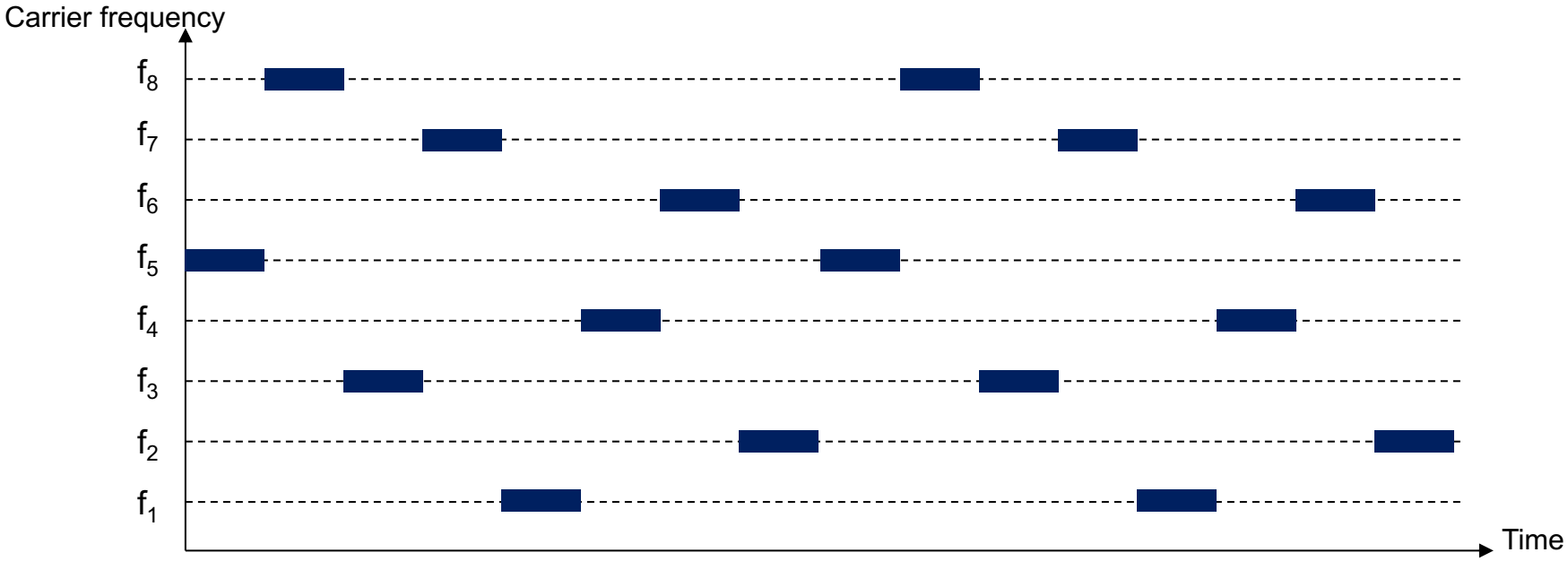
- Two different forms
  - Fast hopping: multiple frequencies per symbol duration T_s, i.e., $T_d < T_s$
  - Slow hopping: multiple symbols per dwell time T_d, i.e., $T_d > T_s$

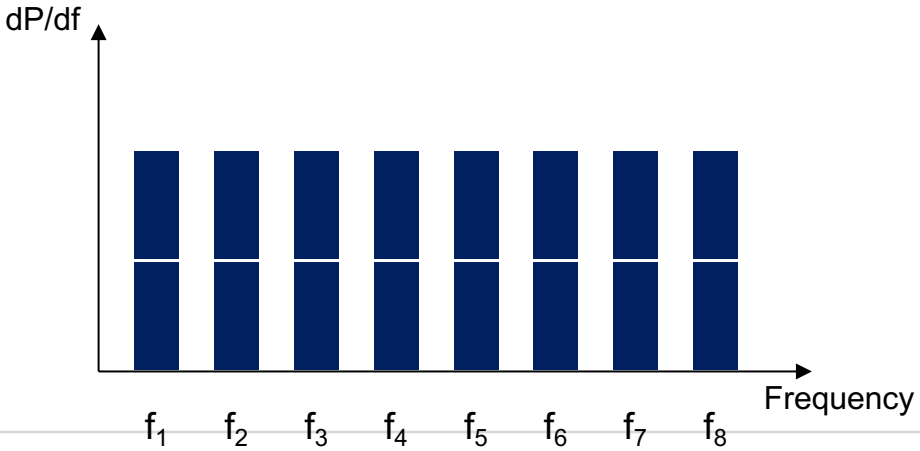
FHSS - Slow vs. fast hopping

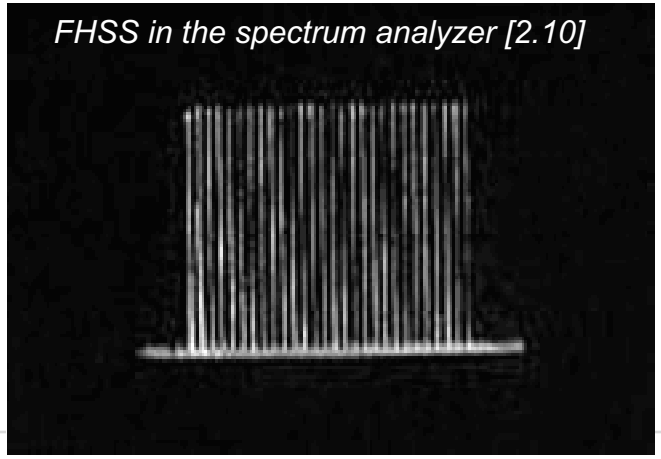


T_s : Symbol duration T_d : Dwell time

FHSS: Power density







Direct Sequence Spread Spectrum (DSSS)

- Application of code multiplexing: Signal spread by multiplying each data bit with bits of a code / chipping sequence
  - Energy per bit is distributed across multiple chips
  - Increases bandwidth (see below)
  - Code bits (or chips) are generated with a given chip rate R_c
  - For a (data) bit rate R_b we have a spreading factor $SF = R_c/R_b$
  - Number of symbols required to transmit a data bit is $N_D = SF$
- Observations:
  - For a fixed chip rate R_c (and modulation rate) data rate R_b decreases with the spreading factor SF: $R_b = R_c/SF$
  - To achieve a fixed data rate R_d, we need to increase chip rate R_c (and modulation rate): $R_c = SF \cdot \mathrm{R_b}$

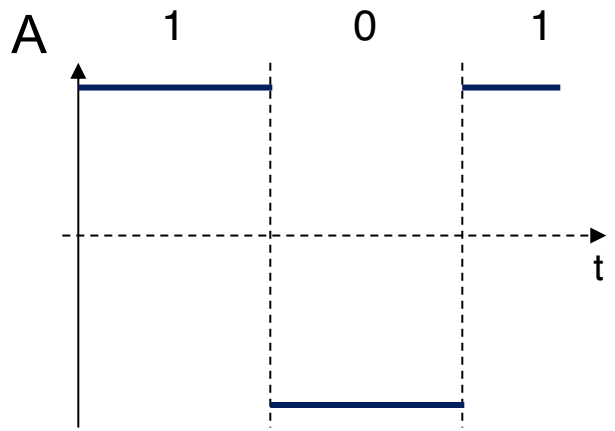
DSSS: Power density (1)

- Assumptions:
  - Input: Bipolar (random) data signal $s(t)$ with symbol duration T and amplitude $\pm A$
  - Transmission by signal v(t) with binary phase modulation (Binary Phase Shift Keying, BPSK) and carrier frequency f_0

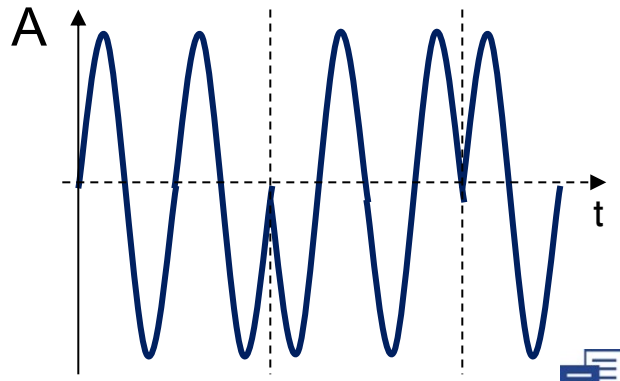
- The power density spectrum of the BPSK signal $v(t)$ is identical to the power density spectrum $L_s(f)$ of the original signal $s(t)$ shifted by f_0 :

$$L_S(f) = A^2 T \left(\frac{\sin \pi f T}{\pi f T} \right)$$

Data signal $s(t)$:



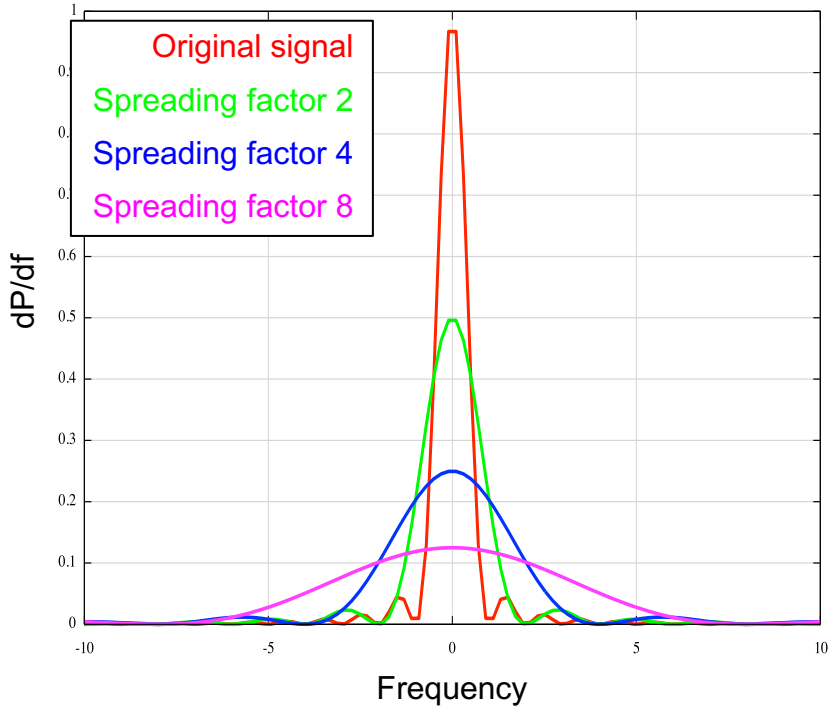
Phase-modulated signal $v(t)$:

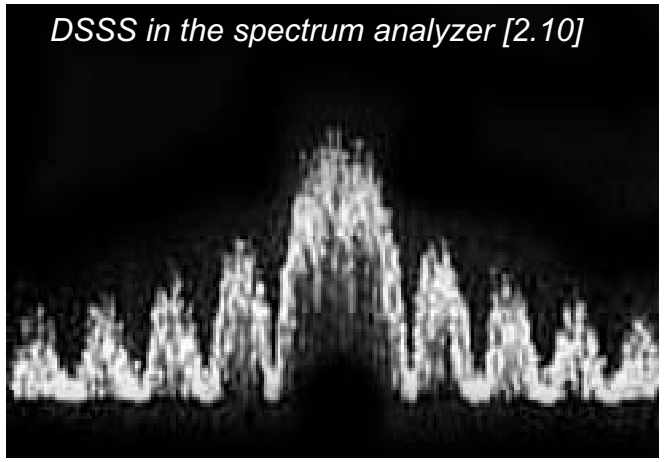


[2.11]

DSSS: Power density (2)

- With DSSS, several chips are modulated instead of one data bit
  - →For fixed data rate, symbol duration T_s must decreases with the spreading factor ("smaller T")
  - →Higher bandwidth of the resulting signal





[2.11]

Comparison FHSS vs. DSSS

FHSS	DSSS
Low costs due to simple transmitter / receiver	Higher costs due to complex transmitters / receivers
Low power consumption	Higher power consumption
Lower data rates	Higher data rates
Lower ranges	Longer ranges
Highest tolerance of interference	Lower tolerance of narrowband interference

[2.9]

Application examples

FHSS	DSSS
• WiFi 802.11 (hardly any products)	• WiFi 802.11 / 802.11b
• Bluetooth	• UMTS (multiple access through Code Division Multiple Access)

[2.13]

Orthogonal frequency division multiplexing (OFDM)

OFDM: The basic idea (1)

- Orthogonal frequency multiplexing is a special case of frequency multiplexing
- Analogy:
  Classic FDM → Tap OFDM → Shower head

One water flow, cannot be subdivided many small water flows

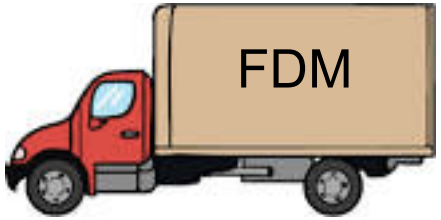
- Advantages:
  - I can stop the tap with my thumb, but not the shower head
  - Each small water flow carries less water

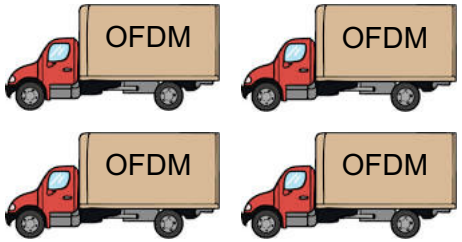
OFDM: The basic idea (2)

- Further analogy: Shipping a (divisible) delivery by truck

- We have the choice:

  One large truck Several small trucks





- Advantages:
  - In the event of an accident, only ¼ of the load is lost
  - If there are several lanes, the trucks can drive in parallel (and keep more distance)

Parallelization of the serial data stream

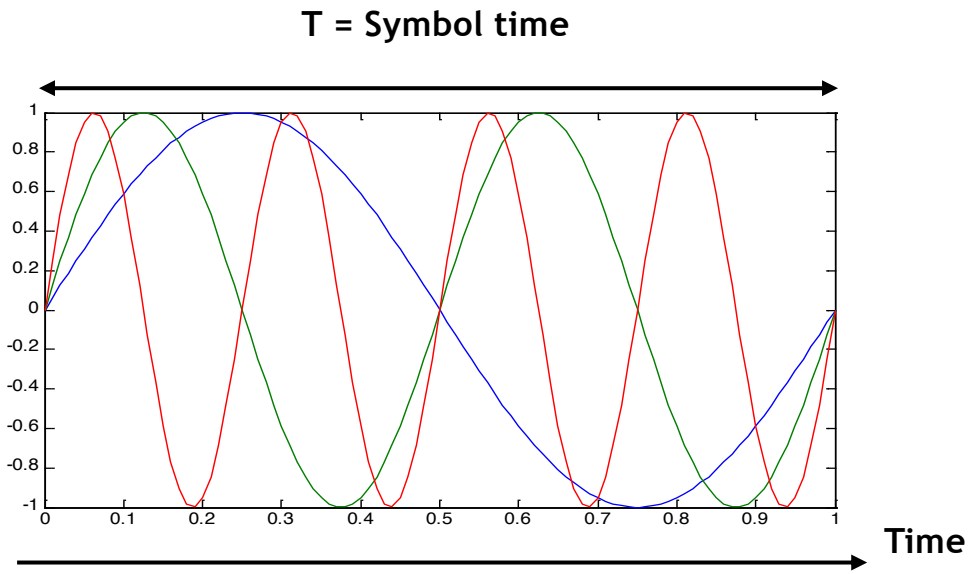
1, 1, -1, -1, 1, 1, 1, -1, 1, -1, -1, -1, -1, 1, -1, -1, -1 , 1, 1, -1, -1, -1, 1, 1

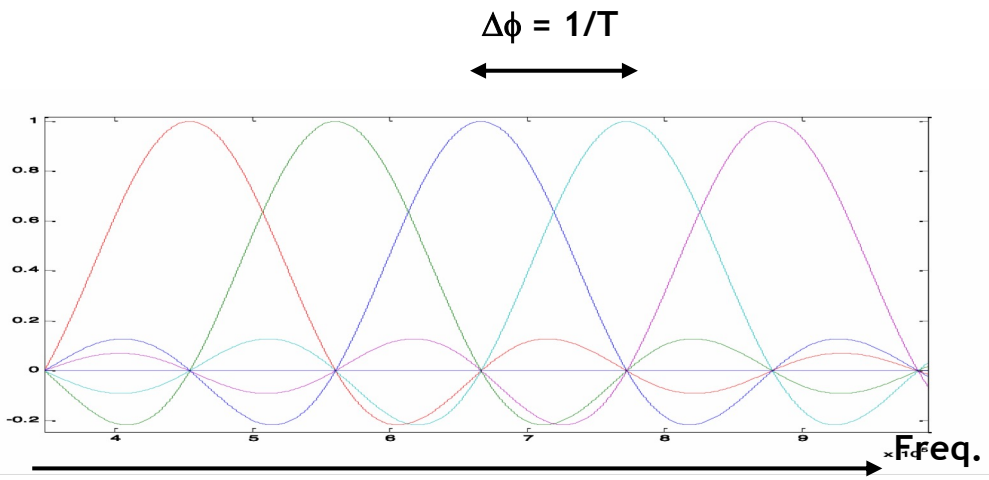
Partial data stream 1	Partial data stream 2	Partial data stream 3	Partial data stream 4
1	1	-1	-1
1	1	1	-1
1	-1	-1	-1
-1	1	-1	-1
-1	1	1	-1
-1	-1	1	1

Open question: How are the partial data streams transferred?

Carrier frequencies in OFDM

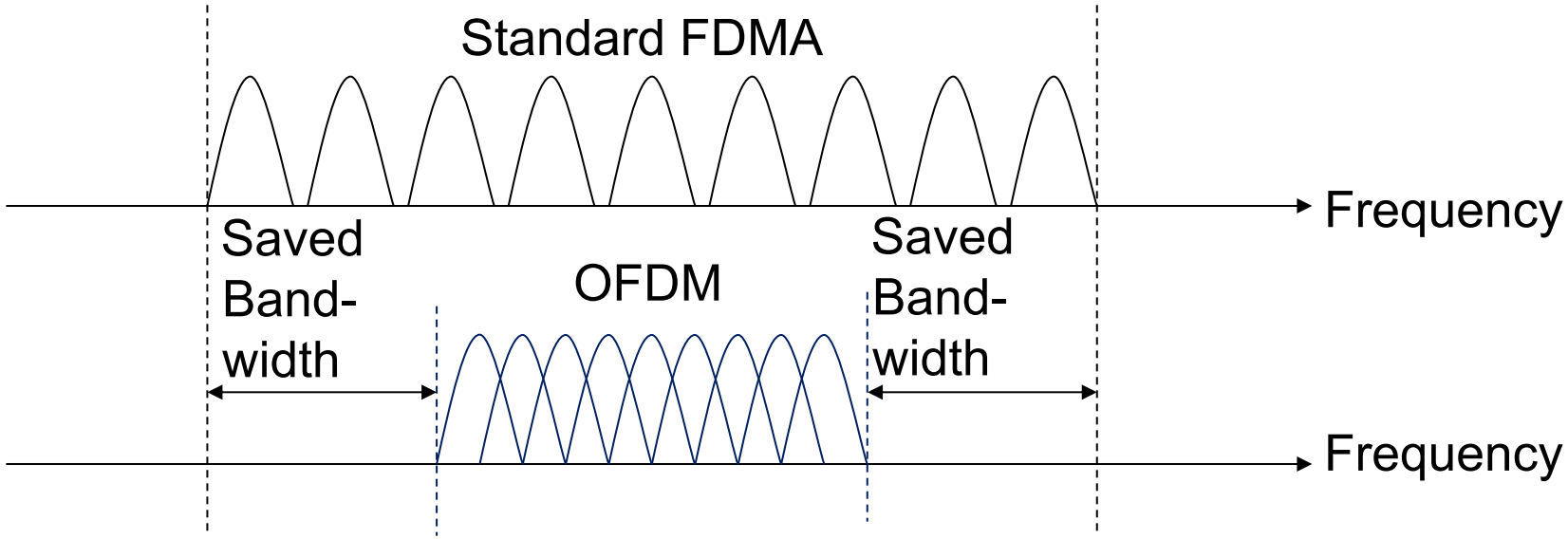
- The carrier frequencies used are harmonics of a fundamental frequency
- Time domain: Simple sinusoidal oscillations
  - Each oscillation has an integer number of cycles above the symbol time T
  - Fundamental frequency $f_0 = 1/T$
  - Further frequencies $f_k = k \cdot f_0$
- → Oscillations are orthogonal
- Frequency domain: Overlapping sinusoidal oscillations
  - Designated as subcarrier
  - Typically very narrow, e.g. 15 kHz in LTE





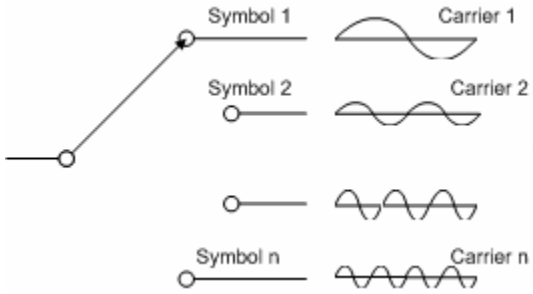
OFDM vs. FDM

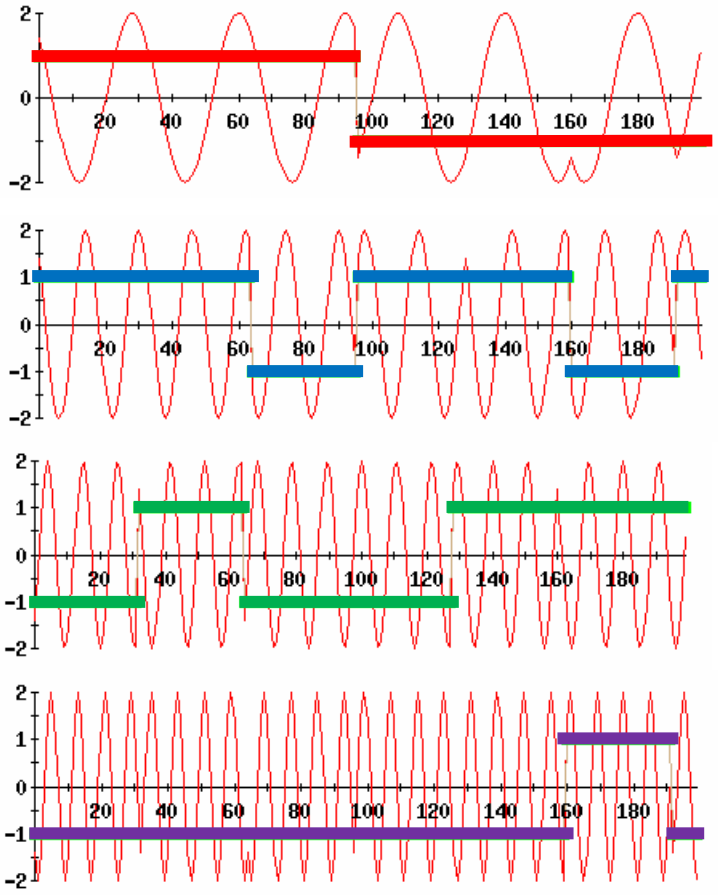
- Difference to traditional FDM:
  Smaller distance between the (subcarrier) frequencies used
  - Relatively large bandwidth required for FDMA due to large distance
  - With OFDM, the subcarrier can even overlap
    - This results in savings in terms of the required bandwidth



Modulation of the partial data streams of the subcarriers

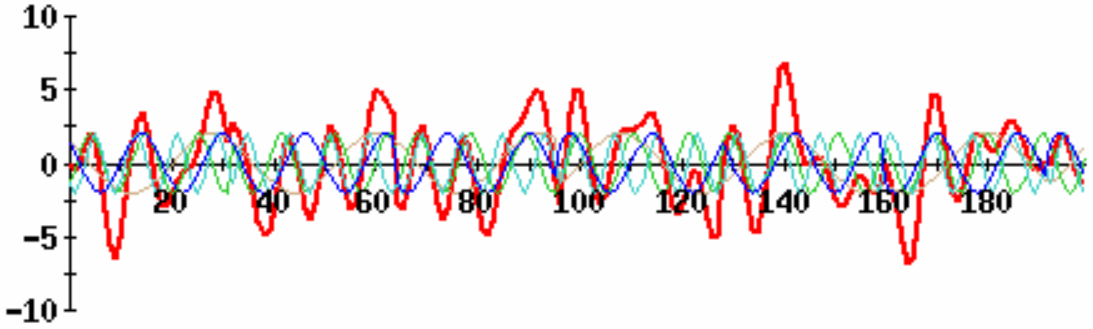
Partial data stream 1	Partial data stream 2	Partial data stream 3	Partial data stream 4
1	1	-1	-1
1	1	1	-1
1	-1	-1	-1
-1	1	-1	-1
-1	1	1	-1
-1	-1	1	1

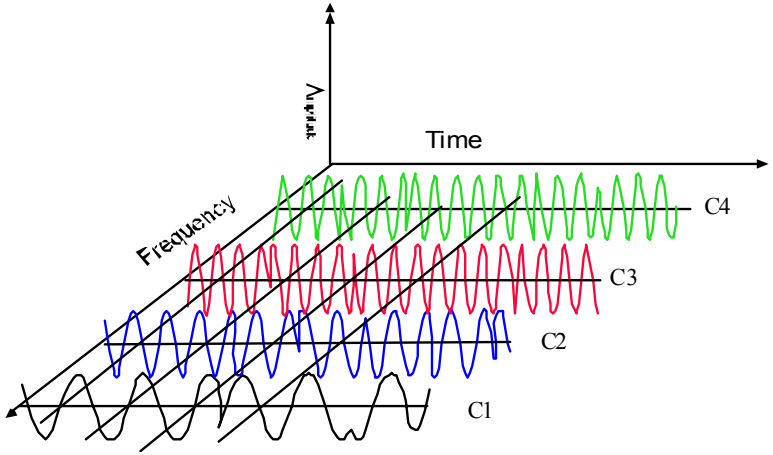




Resulting OFDM signal

- Transmitted signal is the sum of the signals of the subcarriers

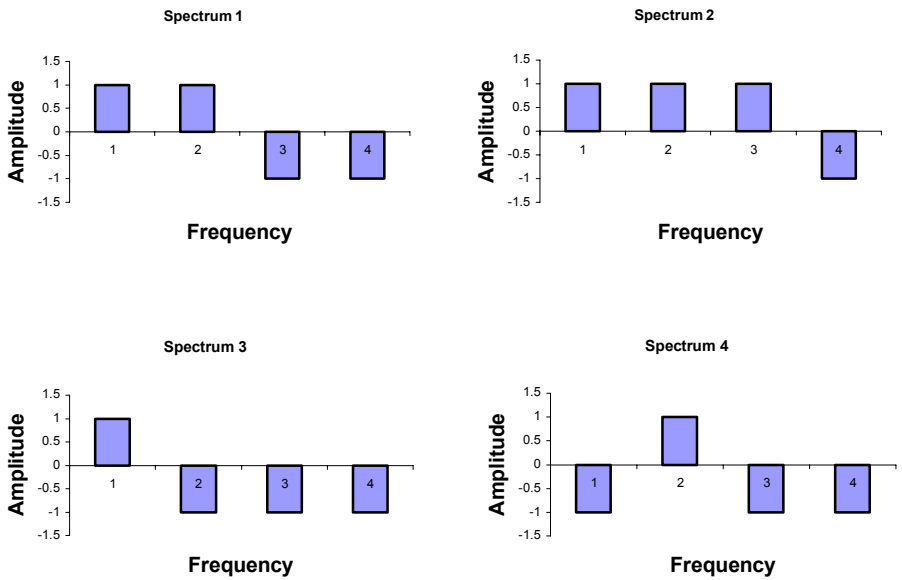




Determination of the OFDM signal

- The incoming (parallel) bit stream can be interpreted as a discrete frequency spectrum with four frequencies

Partial data stream 1	Partial data stream 2	Partial data stream 3	Partial data stream 4
1	1	-1	-1
1	1	1	-1
1	-1	-1	-1
-1	1	-1	-1
-1	1	1	-1
-1	-1	1	1



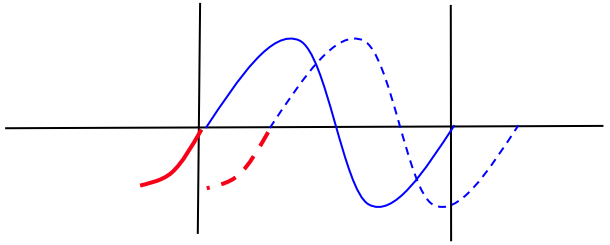
- An inverse fast Fourier transform (IFFT) transforms this from the frequency domain into the time domain

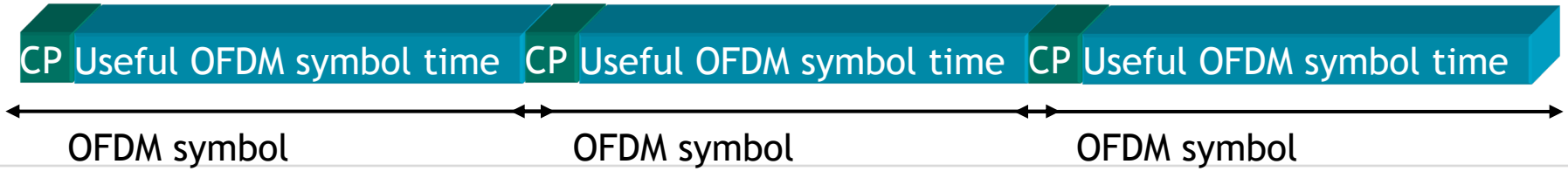
Cyclic prefix: Protection against ISI

- OFDM alone does not protect against inter-symbol interference!
  - ISI (between OFDM symbols) is almost completely eliminated by inserting a guard interval (guard time T_G)

- Open question: What is transmitted during the protection intervals?
  - "Gaps" in signals are technically not possible!

    → Insertion of cyclic prefix (CP)



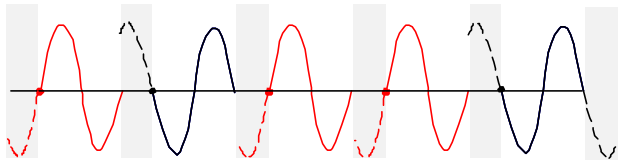


Filling up the protection intervals

- Solution idea 1: "Run symbol longer"

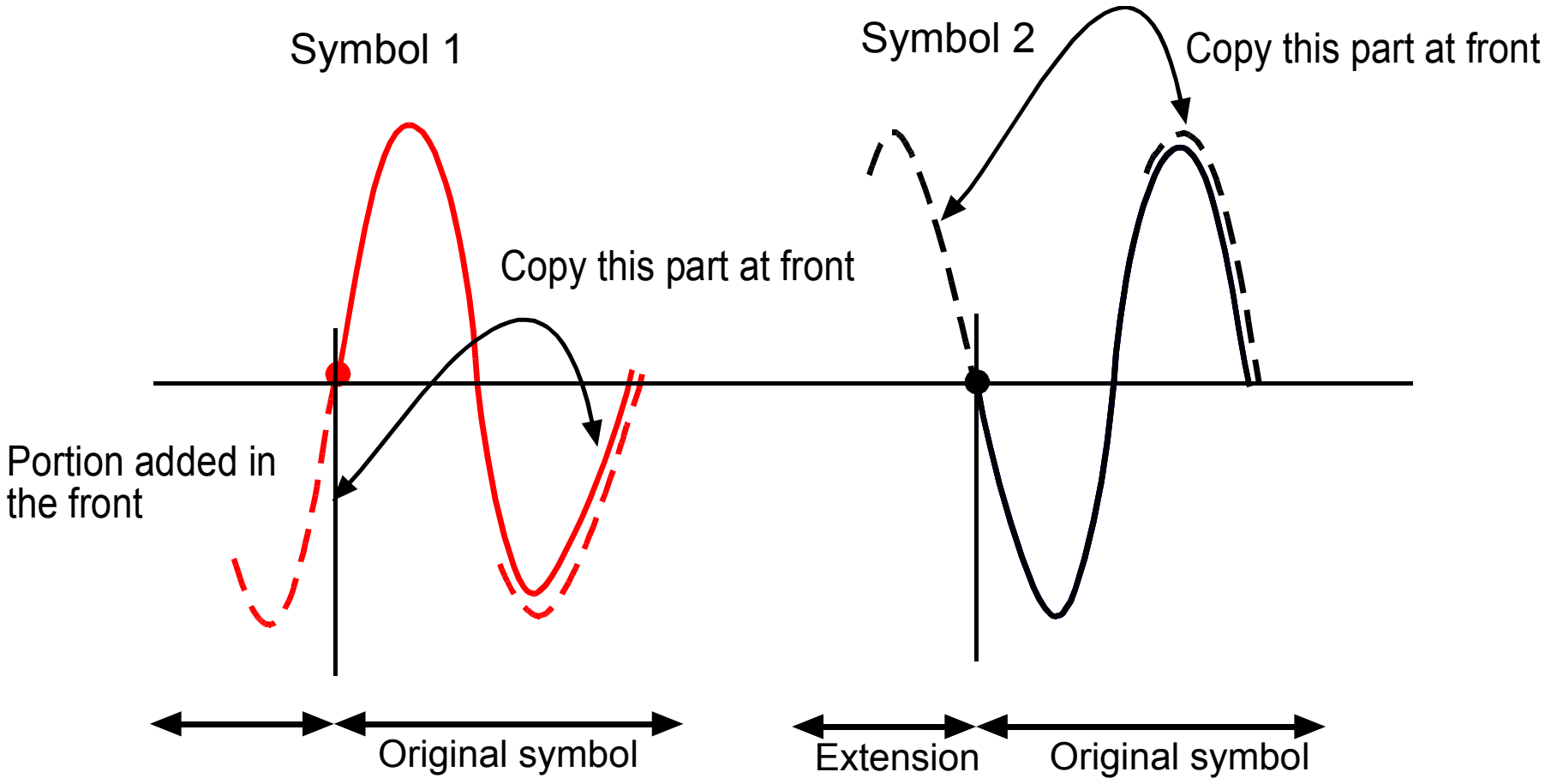
  - Does not help, because ISI just concerns the beginning of the symbol (phase detection)!

- Solution idea 2: "Start symbol earlier"



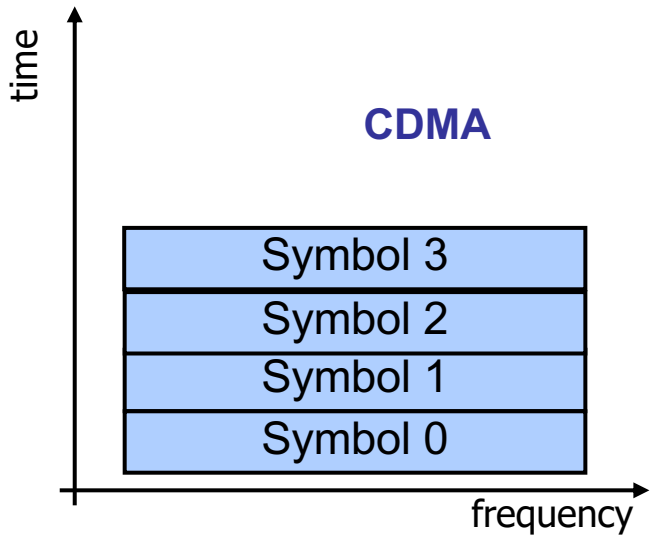
  - Start of symbol is outside ISI range!

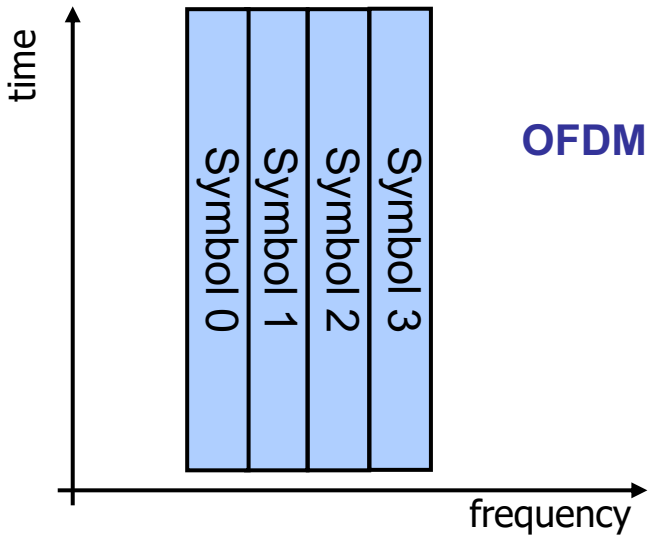
Determination of the cyclic prefix



Comparison OFDM vs. CDMA (1)

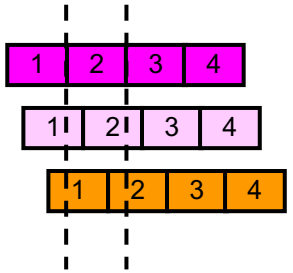
- CDMA: The symbol to be modulated is transmitted over a relatively short symbol time and large bandwidth
  - Example UMTS HSPA: 4.17 µsec symbol time and 3.84 MHz bandwidth
  - To get more bandwidth → Use more CDMA codes
- OFDM: The symbol to be modulated is transmitted over a relatively long symbol time and narrow bandwidth
  - Example LTE: 66.6 µsec symbol time and 15 kHz bandwidth
  - For higher data rates, more symbols can be sent via more subcarriers
    → Increases bandwidth usage





Comparison OFDM vs. CDMA (2)

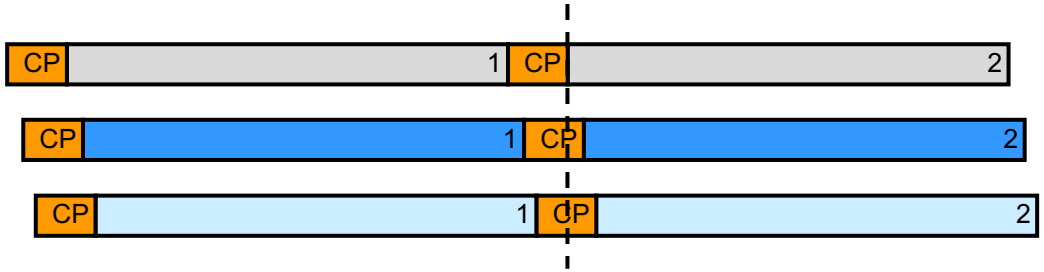
- Short symbol times in CDMA lead to ISI in multipath propagation



CDMA symbols

Multipath reflections of a symbol overlap subsequent symbols → ISI

- Long symbol times in OFDM (together with CP) suppress ISI in multipath propagation



Little to no overlap in symbols

Exercise: CDMA vs. OFDM

We compare a UMTS transmission using CDMA with an LTE transmission using OFDM.

a) UMTS transmits data using CDMA with a chip rate of 3.84 Mchips/s and uses QPSK as the modulation scheme. What is the symbol duration in this case?

b) LTE sends data using OFDM, divided into frames of length 0.5 ms. A frame consists of 7 symbols; each modulated with 64-QAM. What is the symbol duration?

c) How large must the difference in the lengths of two propagation paths between transmitter and receiver be so that a symbol transmitted on the shorter path is exactly overlapped by the previous symbol on the longer path in each case?

Multiple Input Multiple Output (MIMO)

Multiple-Input-Multiple-Output (MIMO) - Motivation

- Channel capacity $\mathbb{C}$
  - Maximum amount of information that can be transmitted error-free at a given bandwidth B
    - Dependent on signal-to-noise ratio $SNR = P_s/P_n$ for average powers of a signal P_s and background noise P_n
  - According to Shannon for channel with additive white Gaussian noise (AWGN)

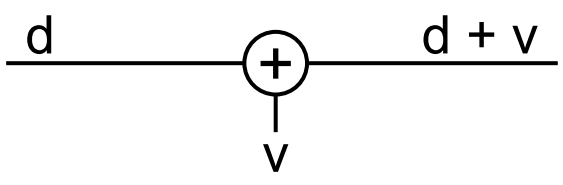
$$\mathbb{C} = B \log_2 (1 + SNR)$$

- Spectral efficiency $C = \mathbb{C}/B$
  - Important optimization criterion in mobile communications (Example: UMTS license fees € 50 billion for 60 MHz)

[2.5]

Capacity gain through parallel transmission (1)

- Single data path
  - $C_1 = \log_2(1 + SNR)$

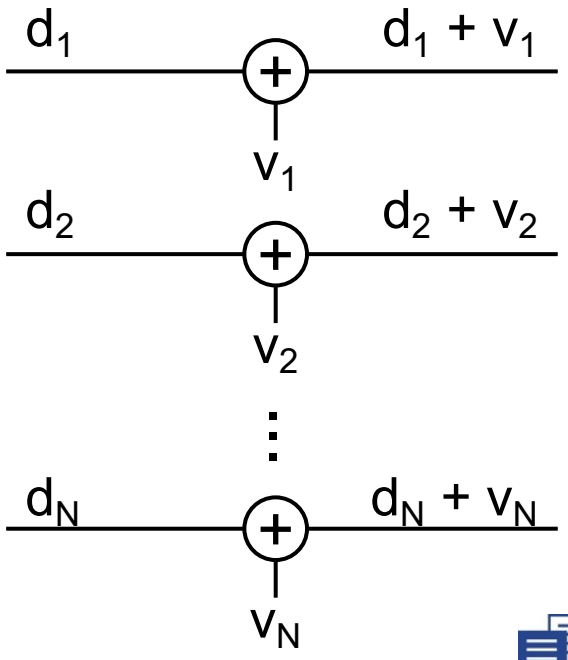


[v = noise]

- N parallel transmissions
  - Transmission power is distributed to all N paths

    N transmission channels...
  - $C_N = N \log_2(1 + SNR/N)$

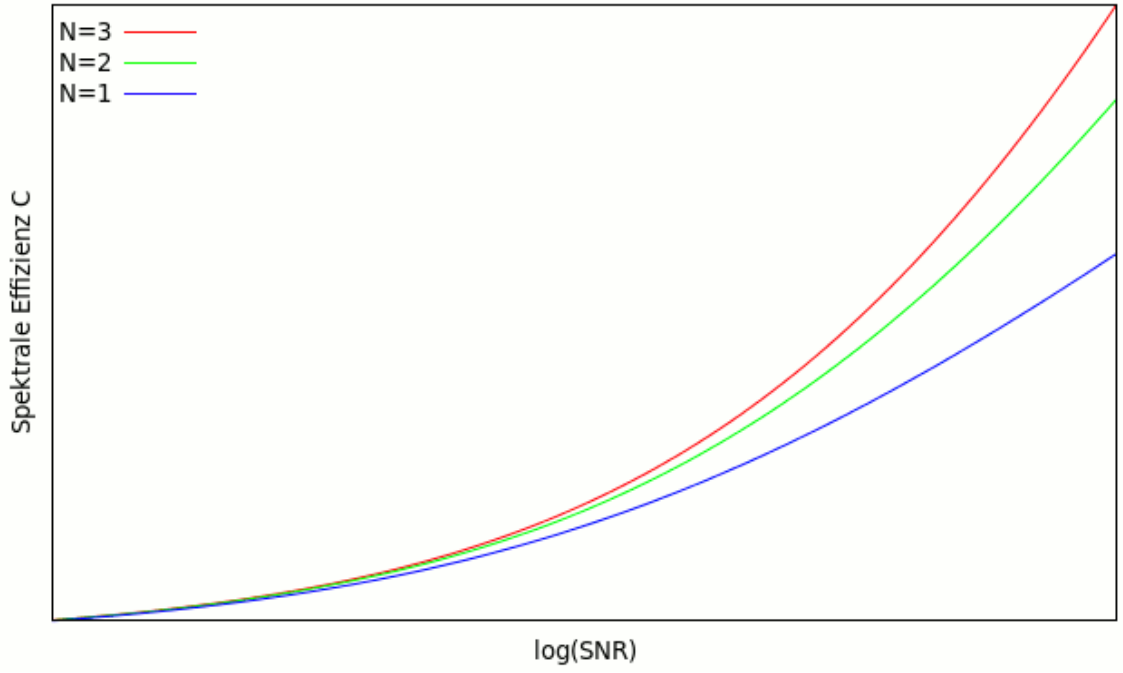
    ... with 1/N transmission power each



[2.5]

Capacity gain through parallel transmission (2)

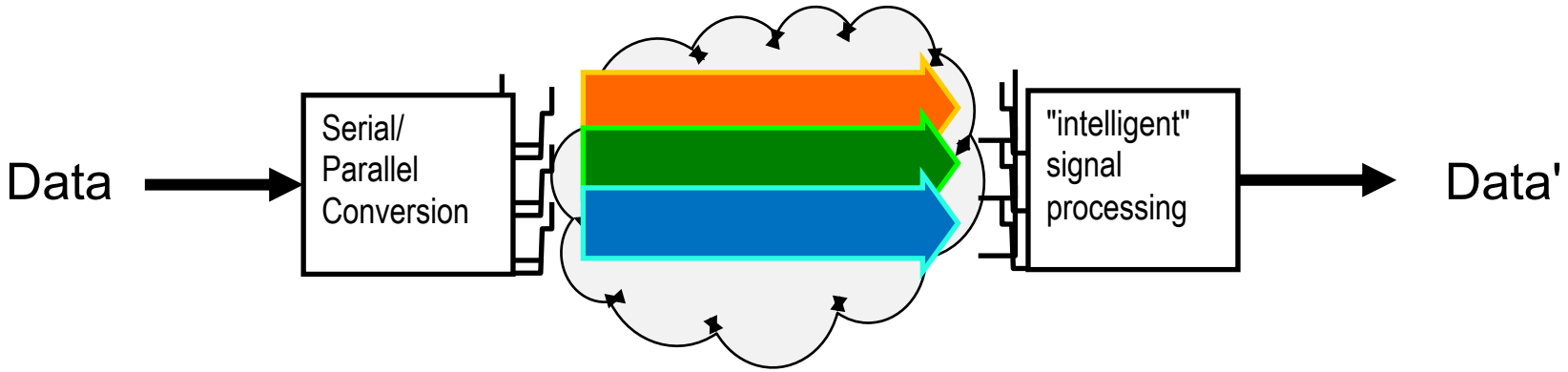
- Significantly higher spectral efficiency with large SNR
  - Interesting for local networks (e.g. WLAN)



[2.5]

MIMO - Application

- Data stream is divided into several partial streams
  - Simultaneous radiation using multi-element antennas
- Receiver detects and separates superimposed signals with intelligent signal processing
  - Generally channel knowledge required (estimates from known pilot signals, statistical properties)
- Application examples
  - WLAN IEEE 802.11n, LTE



MIMO transmission channel

[2.5]

Exercise: MIMO

We consider a transmission between an LTE transmitter and an LTE receiver.

a) Calculate the capacity C for a Single-Input Single-Output (SISO) transmission channel with a signal-to-noise ratio (SNR) of $20dB$.
Note that SNR values in dB must be converted to the linear domain:

$$SNR_{linear} = 10^{\frac{SNR_{dB}}{10}}$$

b) Assume you are using LTE Release 8 and increase the number of parallel transmission channels to $N = 4$ (4x4 MIMO). Calculate the capacity C_N for this case.

c) Interpret the result and compare the capacity between the SISO and MIMO systems. Explain why MIMO provides a significant improvement in spectral efficiency, especially at high SNR.

Summary

- Characteristics of wireless transmissions
  - Ideal: Attenuation with the square of the distance
  - Real: Usually worse
    - Scattering, diffraction, shadowing Reflection
    - Inter-symbol interference, fast and slow fading
- Modulation
  - Representation of symbols by modifying electromagnetic waves
  - One or more user data bits per symbol
    - E.g. 4 bits/symbol for 16-QAM
- Multiplexing
  - Multiple parallel transmissions
  - Dimensions space, time, frequency, code
- Band spreading techniques
  - Reduction of the influence of interference
  - Frequency hopping or direct sequence
- Orthogonal frequency multiplexing is a special case of frequency multiplexing
  - Uses narrow support beams with small spacing
  - Serial data streams are distributed in parallel to the subcarriers, enabling long symbol times

Multiple Input Multiple Output is a special case of room multiplexing

- Uses independent propagation paths

Exercises - Wireless transmission

2.1 What is the relationship between transmit and receive power?
2.2 What is multipath propagation?
2.3 How does intersymbol interference arise and what effects does it have?
2.4 What is the relationship between frequency and wavelength?
2.5 In which frequency ranges are the common mobile communication networks located?
2.6 Which digital modulation techniques do you know?
2.7 What is the function of analog modulation?
2.8 In what ways can signals be displayed?
2.9 What are the advantages of CDMA and how does it work?
2.10 Compare OFDM with frequency division multiplexing.
2.11 What types of spread spectrum techniques are there and how do they work?
do they work?

References and further reading

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